MATA37H3 — Week 06

Numerical Integration: Midpoint, Trapezoidal and Simpson's Rule

Recap and orientation

Weeks 1-5 built the exact toolkit for $\int_a^b f(x)\,dx$: antiderivatives, integration techniques and improper integrals. This week covers the situation where **no elementary antiderivative is available** (for example e^{-x^2}, $\sin(x^2)$, $\dfrac{\sin x}{x}$), or where f is known only from a **table of measured values**. The integral is then approximated to any desired accuracy using only values of f, together with a guaranteed **error bound**.

Why this matters

The three rules differ in how much of the graph they capture on each subinterval: a rectangle (Midpoint), a straight chord (Trapezoidal) or a parabola (Simpson). Knowing the error bounds lets you choose n in advance, and the sign of f'' tells you in which direction each rule errs.

DEFINITION **Setting: Approximating a Definite Integral**

To approximate $\displaystyle\int_a^b f(x)\,dx$, divide $[a,b]$ into n equal subintervals of width

$$\Delta x = \frac{b-a}{n}, \qquad x_i = a + i\,\Delta x, \quad x_0 = a < x_1 < \cdots < x_n = b.$$

All rules use only values of f at points of this grid (or at the midpoints of its subintervals).

KEY FACT **The Midpoint Rule**

$$M_n = \Delta x\Big[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\Big], \qquad \bar{x}_i = \frac{x_{i-1}+x_i}{2} \text{ (midpoint of the } i\text{-th subinterval).}$$

Geometrically, M_n sums the areas of rectangles whose height is the function value at each subinterval's midpoint.

KEY FACT **The Trapezoidal Rule**

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

Geometrically, T_n sums the areas of trapezoids formed by joining consecutive points $(x_{i-1}, f(x_{i-1}))$ and $(x_i, f(x_i))$ with straight lines. Weights: 1 at the two end points, 2 at every interior grid point.

KEY FACT **Error Bounds for M_n and T_n**

If $|f''(x)| \le K$ for all $x \in [a,b]$, then the errors $E_M = \left|\int_a^b f\,dx - M_n\right|$ and $E_T = \left|\int_a^b f\,dx - T_n\right|$ satisfy

$$E_M \le \frac{K(b-a)^3}{24n^2}, \qquad E_T \le \frac{K(b-a)^3}{12n^2}.$$

The Trapezoidal bound is exactly twice the Midpoint bound. For a function of one concavity, M_n and T_n err in opposite directions.

KEY FACT **Simpson's Rule**

For **even** n,

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \cdots + 4f(x_{n-1}) + f(x_n)\Big].$$

Weights: 1 at the end points, then $4, 2, 4, 2, \ldots, 4$ at the interior points (the interior run starts and ends with 4 because n is even). Simpson's Rule fits a parabola through each consecutive triple of grid points, so it integrates every polynomial of degree ≤ 3 **exactly**.

KEY FACT **Error Bound for S_n**

If $\left|f^{(4)}(x)\right| \le M$ for all $x \in [a,b]$, then

$$E_S = \left|\int_a^b f\,dx - S_n\right| \le \frac{M(b-a)^5}{180n^4}.$$

The n^4 (versus n^2 for M_n and T_n) means that doubling n divides the Trapezoidal bound by 4 and the Simpson bound by 16.

KEY FACT **Concavity and the Relationship Between the Rules**

If $f'' > 0$ on $[a,b]$ (concave up), then $M_n \le \int_a^b f\,dx \le T_n$. If $f'' < 0$ (concave down), then $T_n \le \int_a^b f\,dx \le M_n$. Simpson's Rule is built from the two:

$$S_{2n} = \frac{2M_n + T_n}{3}.$$

Worked Examples

EXAMPLE 1

Level: Foundational. Trapezoidal Rule, basic computation

Approximate $\displaystyle\int_0^2 x^2\,dx$ using T_4.

Step 1. Formula used (Trapezoidal Rule, $n = 4$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$T_4 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)\Big].$$

Step 2. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{4} = 0.5.$$

Step 3. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 0 \qquad x_1 = 0.5 \qquad x_2 = 1 \qquad x_3 = 1.5 \qquad x_4 = 2$$

Step 4. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = (0)^2 = 0 \qquad f(x_1) = (0.5)^2 = 0.25$$
$$f(x_2) = (1)^2 = 1 \qquad f(x_3) = (1.5)^2 = 2.25$$
$$f(x_4) = (2)^2 = 4$$

Step 5. Insert the weights $(1, 2, 2, \ldots, 2, 1)$ and the function values:

$$T_4 = \frac{0.5}{2}\Big[\,0 + 2(0.25) + 2(1) + 2(2.25) + 4\,\Big]$$

Step 6. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; 0 + 0.5 + 2 + 4.5 + 4 \; = \; 11$$

Step 7. Multiply by the front factor:

$$T_4 = 0.25 \times 11 = 2.75.$$

Step 8. Exact value for comparison, by the Fundamental Theorem of Calculus:

$$\int_0^2 x^2\,dx = \Big[\frac{x^3}{3}\Big]_0^2 = \frac{2^3}{3} - \frac{0^3}{3} = \frac{8}{3} \approx 2.666667.$$

Step 9. Error:

$$T_4 - I = 2.75 - 2.66667 = 0.0833333, \qquad |I - T_4| = 0.08333.$$

The difference is positive, so T_4 **overestimates** the exact value.

Step 10. Reason from concavity: $f(x) = x^2$ has $f'(x) = 2x$ and $f''(x) = 2 > 0$, so the graph is concave up. On each subinterval the chord joining two points of the graph lies **above** the graph, so each trapezoid has more area than the region under the curve. Hence $T_4 > \displaystyle\int_0^2 x^2\,dx$, which agrees with the computed difference $+0.083333$.

Answer. $T_4 = 2.75$, an overestimate of the exact value $\dfrac{8}{3} \approx 2.6667$ (the graph is concave up).

EXAMPLE 2

Level: Foundational. Midpoint Rule: same integral, opposite sign of error

Approximate $\displaystyle\int_0^2 x^2\,dx$ using M_4.

Step 1. Formula used (Midpoint Rule, $n = 4$):

$$M_n = \Delta x\left[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\right].$$

For $n = 4$ this is

$$M_4 = \Delta x\Big[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) + f(\bar{x}_4)\Big].$$

Step 2. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{4} = 0.5.$$

Step 3. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$:

$$\bar{x}_1 = \frac{0+0.5}{2} = 0.25 \qquad \bar{x}_2 = \frac{0.5+1}{2} = 0.75$$

$$\bar{x}_3 = \frac{1+1.5}{2} = 1.25 \qquad \bar{x}_4 = \frac{1.5+2}{2} = 1.75$$

Step 4. Function values at the midpoints (rounded to 6 significant figures where necessary):

$$f(\bar{x}_1) = (0.25)^2 = 0.0625 \qquad f(\bar{x}_2) = (0.75)^2 = 0.5625$$

$$f(\bar{x}_3) = (1.25)^2 = 1.5625 \qquad f(\bar{x}_4) = (1.75)^2 = 3.0625$$

Step 5. Add the 4 midpoint values (all weights are 1):

$$\text{bracket} = \ 0.0625 + 0.5625 + 1.5625 + 3.0625 \ = \ 5.25$$

Step 6. Multiply by Δx:

$$M_4 = 0.5 \times 5.25 = 2.625.$$

Step 7. Error:

$$M_4 - I = 2.625 - 2.66667 = -0.0416667, \qquad |I - M_4| = 0.04167.$$

The difference is negative, so M_4 **underestimates** the exact value.

Step 8. Reason from concavity: on a subinterval with midpoint $\bar{x}$, the rectangle of height $f(\bar{x})$ has the same area as the trapezoid under the **tangent line** at $\bar{x}$ (the tangent line has average height $f(\bar{x})$ over a symmetric interval). For a concave-up function the graph lies above its tangent lines, so the area under the curve is larger than the rectangle area. Hence $M_4 < \displaystyle\int_0^2 x^2\,dx$, in agreement with the sign -0.041667 found above.

Step 9. Compare the sizes of the two errors from Example 1 and this example:

$$|I - T_4| = 2.75 - \frac{8}{3} = 0.083333, \qquad |I - M_4| = \frac{8}{3} - 2.625 = 0.041667,$$

and $\dfrac{0.083333}{0.041667} = 2$, so the Trapezoidal error is twice the Midpoint error here, as the bounds $\dfrac{K(b-a)^3}{12n^2}$ and $\dfrac{K(b-a)^3}{24n^2}$ (ratio 2) suggest.

Answer. $M_4 = 2.625$, an underestimate of $\dfrac{8}{3} \approx 2.6667$; $|I - T_4| = 2\,|I - M_4|$.

EXAMPLE 3

Level: Foundational. Simpson's Rule: exact for low-degree polynomials

Approximate $\displaystyle\int_0^2 x^2\,dx$ using S_4.

Step 1. Formula used (Simpson's Rule, $n = 4$):

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$S_4 = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)\Big].$$

Step 2. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{4} = 0.5.$$

Step 3. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 0 \qquad x_1 = 0.5 \qquad x_2 = 1 \qquad x_3 = 1.5 \qquad x_4 = 2$$

Step 4. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = (0)^2 = 0 \qquad f(x_1) = (0.5)^2 = 0.25$$
$$f(x_2) = (1)^2 = 1 \qquad f(x_3) = (1.5)^2 = 2.25$$
$$f(x_4) = (2)^2 = 4$$

Step 5. Insert the weights $(1, 4, 2, 4, \ldots, 2, 4, 1)$ and the function values:

$$S_4 = \frac{0.5}{3}\Big[\,0 + 4(0.25) + 2(1) + 4(2.25) + 4\,\Big]$$

Step 6. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; 0 + 1 + 2 + 9 + 4 \; = \; 16$$

Step 7. Multiply by the front factor:

$$S_4 = \frac{0.5}{3} \times 16 = \frac{16}{6} = \frac{8}{3} \approx 2.666667.$$

Step 8. Compare with the exact value $\displaystyle\int_0^2 x^2\,dx = \frac{8}{3}$ (Example 1): the two agree, so S_4 has **zero error**.

Step 9. Two explanations of the zero error. (i) Simpson's Rule fits a parabola through each three consecutive grid points and integrates the parabola exactly; $f(x) = x^2$ already is a parabola, so each piece is reproduced exactly (the same holds for every polynomial of degree ≤ 3). (ii) The error bound is $\dfrac{M(b-a)^5}{180n^4}$ with $|f^{(4)}(x)| \leq M$; here $f'(x) = 2x$, $f''(x) = 2$, $f^{(3)}(x) = 0$, $f^{(4)}(x) = 0$, so $M = 0$ and the error bound is 0.

Answer. $S_4 = \dfrac{8}{3}$, exactly (no error).

EXAMPLE 4

Level: Intermediate. Error bound for the Trapezoidal Rule

Find a bound on the error when approximating $\displaystyle\int_1^2 \frac{1}{x}\,dx$ using T_{10}.

Step 1. Identify the pieces: $f(x) = \dfrac{1}{x} = x^{-1}$, $a = 1$, $b = 2$, $n = 10$, and the Trapezoidal bound is

$$E_T \le \frac{K(b-a)^3}{12n^2}, \qquad K \text{ with } |f''(x)| \le K \text{ on } [1,2].$$

Step 2. Differentiate twice: $f'(x) = -x^{-2} = -\dfrac{1}{x^2}$ and $f''(x) = 2x^{-3} = \dfrac{2}{x^3}$.

Step 3. Find the maximum of $|f''|$ on $[1,2]$. On this interval $x > 0$, so $|f''(x)| = \dfrac{2}{x^3}$. Its derivative is $f^{(3)}(x) = -6x^{-4} = -\dfrac{6}{x^4} < 0$, so $|f''|$ is **decreasing** on $[1,2]$ and its maximum is at the left endpoint $x = 1$:

$$K = |f''(1)| = \frac{2}{1^3} = 2.$$

(At the other end, $|f''(2)| = \dfrac{2}{8} = 0.25$, which is smaller.)

Step 4. Substitute $K = 2$, $b - a = 2 - 1 = 1$, $n = 10$:

$$E_T \le \frac{2 \cdot 1^3}{12 \cdot 10^2} = \frac{2}{12 \cdot 100} = \frac{2}{1200}.$$

Step 5. Simplify: $\dfrac{2}{1200} = \dfrac{1}{600} \approx 0.001667$. So $|\ln 2 - T_{10}| \le \dfrac{1}{600} \approx 0.00167$.

Step 6. Reference check (computed independently): $T_{10} = 0.693771$ and $\ln 2 = 0.693147$, so the actual error is $0.000624 \le 0.001667$, consistent with the bound.

Answer. $E_T \le \dfrac{1}{600} \approx 0.00167$ (with $K = 2$).

EXAMPLE 5

Level: Intermediate. Choosing n in advance for Simpson's Rule

How large must (even) n be so that S_n approximates $\displaystyle\int_0^1 \sin x\,dx$ with error less than 0.0001?

Step 1. Identify the pieces: $f(x) = \sin x$, $a = 0$, $b = 1$, and the Simpson error bound

$$E_S \le \frac{M(b-a)^5}{180n^4}, \qquad |f^{(4)}(x)| \le M \text{ on } [0,1].$$

We want $E_S < 0.0001$ and n even.

Step 2. Differentiate four times: $f'(x) = \cos x$, $f''(x) = -\sin x$, $f^{(3)}(x) = -\cos x$, $f^{(4)}(x) = \sin x$.

Step 3. Bound $|f^{(4)}(x)| = |\sin x|$: for every real x, $|\sin x| \le 1$, so we may take $M = 1$.

Step 4. Require the bound to be below the tolerance:

$$\frac{1 \cdot (1-0)^5}{180n^4} < 0.0001 \implies \frac{1}{180n^4} < 0.0001.$$

Step 5. Multiply both sides by $n^4 > 0$ and divide by 0.0001:

$$n^4 > \frac{1}{180 \times 0.0001} = \frac{1}{0.018} = 55.5556.$$

Step 6. Take the fourth root (positive numbers, so the inequality direction is preserved):

$$n > (55.5556)^{1/4} \approx 2.7298.$$

Check: $2.7298^2 = 7.4518$ and $7.4518^2 = 55.53$.

Step 7. n must be an **even integer** greater than 2.7298. The even integers are $2, 4, 6, \ldots$; $n = 2$ is not greater than 2.7298, so the smallest choice is $n = 4$.

Step 8. Verify $n = 4$: $\dfrac{1}{180 \cdot 4^4} = \dfrac{1}{180 \cdot 256} = \dfrac{1}{46080} = 2.1701 \times 10^{-5} < 10^{-4}$. Verify $n = 2$ fails: $\dfrac{1}{180 \cdot 2^4} = \dfrac{1}{2880} = 3.4722 \times 10^{-4} > 10^{-4}$.

Step 9. Reference check: $S_4 = 0.459708$ and $\displaystyle\int_0^1 \sin x \, dx = 1 - \cos 1 = 0.459698$, so the actual error is 0.0000101, far below 0.0001.

Answer. $n = 4$.

EXAMPLE 6

Level: Intermediate. Numerical integration from a table of data

Water flows into a tank at a rate $r(t)$ litres per minute, measured every 5 minutes:

t (min)	0	5	10	15	20	25	30
$r(t)$	12	15	19	22	20	17	13

Estimate the total volume $\displaystyle\int_0^{30} r(t)\, dt$ using the Trapezoidal Rule T_6, Simpson's Rule S_6 and the Midpoint Rule M_3 (with $n = 3$ subintervals of width 10).

Step 1. Set up the integral. The rate $r(t)$ (litres per minute) is known only at $t = 0, 5, 10, \ldots, 30$ minutes (table above), so the total volume $V = \displaystyle\int_0^{30} r(t)\, dt$ (litres) cannot be found by an antiderivative; use the rules with the tabulated values. The data are equally spaced with spacing 5, so the grid is $x_i = t_i$ with $n = 6$.

Step 2. Trapezoidal Rule with $n = 6$.

Step 3. Formula used (Trapezoidal Rule, $n = 6$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 6$ this is

$$T_6 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_5) + f(x_6)\Big].$$

Step 4. Width of each subinterval:

$$\Delta x = \frac{b - a}{n} = \frac{30 - 0}{6} = 5.$$

Step 5. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 6$:

$$x_0 = 0 \qquad x_1 = 5 \qquad x_2 = 10 \qquad x_3 = 15$$
$$x_4 = 20 \qquad x_5 = 25 \qquad x_6 = 30$$

Step 6. Function values at the grid points:

$$f(x_0) = r(0) = 12 \qquad f(x_1) = r(5) = 15$$
$$f(x_2) = r(10) = 19 \qquad f(x_3) = r(15) = 22$$
$$f(x_4) = r(20) = 20 \qquad f(x_5) = r(25) = 17$$
$$f(x_6) = r(30) = 13$$

Step 7. Insert the weights $(1, 2, 2, \dots, 2, 1)$ and the function values:

$$T_6 = \frac{5}{2}\Big[\, 12 + 2(15) + 2(19) + 2(22) + 2(20) + \; 2(17) + 13 \,\Big]$$

Step 8. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; 12 + 30 + 38 + 44 + 40 + 34 + \; 13 \; = \; 211$$

Step 9. Multiply by the front factor:

$$T_6 = 2.5 \times 211 = 527.5.$$

Step 10. Simpson's Rule with $n = 6$ (even, so allowed; same data).

Step 11. Formula used (Simpson's Rule, $n = 6$):

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big].$$

For $n = 6$ this is

$$S_6 = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + 4f(x_5) + f(x_6)\Big].$$

Step 12. Width of each subinterval:

$$\Delta x = \frac{b - a}{n} = \frac{30 - 0}{6} = 5.$$

Step 13. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \dots, 6$:

$$x_0 = 0 \qquad x_1 = 5 \qquad x_2 = 10 \qquad x_3 = 15$$
$$x_4 = 20 \qquad x_5 = 25 \qquad x_6 = 30$$

Step 14. Function values at the grid points:

$$f(x_0) = r(0) = 12 \qquad f(x_1) = r(5) = 15$$
$$f(x_2) = r(10) = 19 \qquad f(x_3) = r(15) = 22$$
$$f(x_4) = r(20) = 20 \qquad f(x_5) = r(25) = 17$$
$$f(x_6) = r(30) = 13$$

Step 15. Insert the weights $(1, 4, 2, 4, \dots, 2, 4, 1)$ and the function values:

$$S_6 = \frac{5}{3}\Big[\, 12 + 4(15) + 2(19) + 4(22) + 2(20) + \; 4(17) + 13 \,\Big]$$

Step 16. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; 12 + 60 + 38 + 88 + 40 + 68 + \; 13 \; = \; 319$$

Step 17. Multiply by the front factor:

$$S_6 = \frac{5}{3}\,(\approx 1.66667) \times 319 = 531.667.$$

Step 18. Midpoint Rule. The midpoints of subintervals of width 5 are $2.5, 7.5, \dots$, which are **not** in the table. Use subintervals of width 10 instead: $n = 3$, intervals $[0, 10], [10, 20], [20, 30]$, whose midpoints $5, 15, 25$ **are** in the table.

Step 19. Formula used (Midpoint Rule, $n = 3$):

$$M_n = \Delta x \left[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n) \right].$$

For $n = 3$ this is

$$M_3 = \Delta x \Big[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) \Big].$$

Step 20. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{30-0}{3} = 10.$$

Step 21. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$:

$$\bar{x}_1 = \frac{0+10}{2} = 5 \qquad \bar{x}_2 = \frac{10+20}{2} = 15$$

$$\bar{x}_3 = \frac{20+30}{2} = 25$$

Step 22. Function values at the midpoints:

$$f(\bar{x}_1) = r(5) = 15 \qquad f(\bar{x}_2) = r(15) = 22$$

$$f(\bar{x}_3) = r(25) = 17$$

Step 23. Add the 3 midpoint values (all weights are 1):

$$\text{bracket} = \ 15 + 22 + 17 \ = \ 54$$

Step 24. Multiply by Δx:

$$M_3 = 10 \times 54 = 540.$$

Step 25. Summary: $T_6 = 527.5$ litres, $S_6 = 531.7$ litres, $M_3 = 540$ litres. The three estimates agree to within about 2.4%; with only tabulated data there is no formula for f'' or $f^{(4)}$, so an error bound cannot be computed and Simpson's Rule (which uses the most information per subinterval) is taken as the best single estimate.

Answer. $T_6 = 527.5$ L, $S_6 = 531.667$ L, $M_3 = 540$ L.

EXAMPLE 7

Level: Advanced. Simpson's Rule with its error bound

Approximate $\displaystyle\int_0^2 \frac{1}{1+x}\,dx$ using S_6, compare with the exact value, and find the Simpson error bound for $n = 6$.

Step 1. Formula used (Simpson's Rule, $n = 6$):

$$S_n = \frac{\Delta x}{3} \Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n) \Big].$$

For $n = 6$ this is

$$S_6 = \frac{\Delta x}{3} \Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + 4f(x_5) + f(x_6) \Big].$$

Step 2. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{6} = \frac{1}{3} \approx 0.333333.$$

Step 3. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 6$:

$$x_0 = 0 \qquad x_1 = \frac{1}{3} \approx 0.333333 \qquad x_2 = \frac{2}{3} \approx 0.666667 \qquad x_3 = 1$$

$$x_4 = \frac{4}{3} \approx 1.33333 \qquad x_5 = \frac{5}{3} \approx 1.66667 \qquad x_6 = 2$$

Step 4. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0)=\frac{1}{1+0}=1 \qquad f(x_1)=\frac{1}{1+\frac{1}{3}}=0.75$$

$$f(x_2)=\frac{1}{1+\frac{2}{3}}=0.6 \qquad f(x_3)=\frac{1}{1+1}=0.5$$

$$f(x_4)=\frac{1}{1+\frac{4}{3}}=0.428571 \qquad f(x_5)=\frac{1}{1+\frac{5}{3}}=0.375$$

$$f(x_6)=\frac{1}{1+2}=0.333333$$

Step 5. Insert the weights $(1,4,2,4,\dots,2,4,1)$ and the function values:

$$S_6=\frac{1}{9}\Big[\,1+4(0.75)+2(0.6)+4(0.5)+2(0.428571) \\ +\;4(0.375)+0.333333\,\Big]$$

Step 6. Do each multiplication inside the bracket and add the results:

$$\text{bracket}=\;1+3+1.2+2+0.857142+1.5 \\ +\;0.333333\;=\;9.890475$$

Step 7. Multiply by the front factor:

$$S_6=\frac{1}{9}\,(\approx 0.111111)\times 9.890475=1.09894.$$

Step 8. Exact value for comparison: $\displaystyle\int_0^2\frac{dx}{1+x}=\Big[\ln|1+x|\Big]_0^2=\ln 3-\ln 1=\ln 3\approx 1.09861,$ so the actual error is $|\ln 3-S_6|=|1.09861-1.09894|=0.000329$ (S_6 is slightly larger).

Step 9. Now the error **bound**. Write $f(x)=(1+x)^{-1}$ and differentiate four times:

$$f'(x)=-(1+x)^{-2},\quad f''(x)=2(1+x)^{-3},\quad f^{(3)}(x)=-6(1+x)^{-4},\quad f^{(4)}(x)=24(1+x)^{-5}=\frac{24}{(1+x)^5}.$$

Step 10. Maximise $|f^{(4)}|$ on $[0,2]$: for $x\ge 0$ we have $1+x\ge 1$, so $f^{(4)}(x)=\dfrac{24}{(1+x)^5}>0$ and it is a decreasing function of x (the denominator grows). The maximum is at $x=0$:

$$M=f^{(4)}(0)=\frac{24}{1^5}=24.$$

Step 11. Insert into $E_S\le\dfrac{M(b-a)^5}{180n^4}$ with $b-a=2$, $n=6$:

$$E_S\le\frac{24\cdot 2^5}{180\cdot 6^4}=\frac{24\cdot 32}{180\cdot 1296}=\frac{768}{233280}=0.0032922.$$

Step 12. Check: the actual error 0.000329 is smaller than the guaranteed bound 0.003292, as it must be.

Answer. $S_6\approx 1.09894$; exact value $\ln 3\approx 1.09861$; $E_S\le 0.003292$.

EXAMPLE 8

Level: Advanced. Bracketing an integral using concavity

Use the sign of f'' to decide whether T_4 and M_4 over- or under-estimate $\displaystyle\int_0^1 e^x\,dx$, compute both, and state the interval that must contain the integral.

Step 1. Plan: decide the direction of each error from the sign of f'', then compute both rules. For $f(x) = e^x$: $f'(x) = e^x$ and $f''(x) = e^x > 0$ for every x, so the graph is concave up on $[0, 1]$.

Step 2. Concave up means: every trapezoid lies above the curve, so $T_4 \geq \displaystyle\int_0^1 e^x\,dx$; every midpoint rectangle has the area of a tangent-line trapezoid lying below the curve, so $M_4 \leq \displaystyle\int_0^1 e^x\,dx$. Hence

$$M_4 \leq \textstyle\int_0^1 e^x\,dx \leq T_4.$$

Step 3. Compute T_4.

Step 4. Formula used (Trapezoidal Rule, $n = 4$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$T_4 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)\Big].$$

Step 5. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{4} = 0.25.$$

Step 6. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 0 \qquad x_1 = 0.25 \qquad x_2 = 0.5 \qquad x_3 = 0.75 \qquad x_4 = 1$$

Step 7. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = e^0 = 1 \qquad f(x_1) = e^{0.25} = 1.28403$$
$$f(x_2) = e^{0.5} = 1.64872 \qquad f(x_3) = e^{0.75} = 2.117$$
$$f(x_4) = e^1 = 2.71828$$

Step 8. Insert the weights $(1, 2, 2, \ldots, 2, 1)$ and the function values:

$$T_4 = \frac{0.25}{2}\Big[\,1 + 2(1.28403) + 2(1.64872) + 2(2.117) + 2.71828\,\Big]$$

Step 9. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \ 1 + 2.56806 + 3.29744 + 4.234 + 2.71828 \ = \ 13.81778$$

Step 10. Multiply by the front factor:

$$T_4 = 0.125 \times 13.81778 = 1.72722.$$

Step 11. Compute M_4.

Step 12. Formula used (Midpoint Rule, $n = 4$):

$$M_n = \Delta x\,\Big[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\Big].$$

For $n = 4$ this is

$$M_4 = \Delta x\Big[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) + f(\bar{x}_4)\Big].$$

Step 13. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{4} = 0.25.$$

Step 14. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$:

$$\bar{x}_1 = \frac{0 + 0.25}{2} = 0.125 \qquad \bar{x}_2 = \frac{0.25 + 0.5}{2} = 0.375$$
$$\bar{x}_3 = \frac{0.5 + 0.75}{2} = 0.625 \qquad \bar{x}_4 = \frac{0.75 + 1}{2} = 0.875$$

Step 15. Function values at the midpoints (rounded to 6 significant figures where necessary):

$$f(\bar{x}_1) = e^{0.125} = 1.13315 \qquad f(\bar{x}_2) = e^{0.375} = 1.45499$$
$$f(\bar{x}_3) = e^{0.625} = 1.86825 \qquad f(\bar{x}_4) = e^{0.875} = 2.39888$$

Step 16. Add the 4 midpoint values (all weights are 1):

$$\text{bracket} = 1.13315 + 1.45499 + 1.86825 + 2.39888 = 6.85527$$

Step 17. Multiply by Δx:

$$M_4 = 0.25 \times 6.85527 = 1.71382.$$

Step 18. Conclusion:

$$1.71382 \le \int_0^1 e^x \, dx \le 1.72722,$$

so the integral lies in an interval of length 0.0134 found without any antiderivative.

Step 19. Check against the exact value $\displaystyle\int_0^1 e^x \, dx = e - 1 = 1.71828$: indeed $1.71382 < 1.71828 < 1.72722$. Check against the error bounds with $K = e$: $E_T \le \dfrac{e}{12 \cdot 16} = 0.01416$ and $E_M \le \dfrac{e}{24 \cdot 16} = 0.007079$; the actual errors 0.00894 and 0.00446 satisfy them.

Answer. $M_4 \approx 1.7138 \le \displaystyle\int_0^1 e^x dx \le T_4 \approx 1.7272$ (T_4 over, M_4 under).

Practice Problems

Worked-solutions version: every step of each solution is shown in green beneath its problem.

Midpoint and Trapezoidal Rules

1. Approximate $\displaystyle\int_0^4 \sqrt{x}\,dx$ using T_4 (grid points $0, 1, 2, 3, 4$; use $\sqrt{0} = 0,\ \sqrt{1} = 1,\ \sqrt{2} \approx 1.414,\ \sqrt{3} \approx 1.732,\ \sqrt{4} = 2$).

SOLUTION

Step 1. Setup: $f(x) = \sqrt{x}$ on $[0, 4]$ with $n = 4$; the given approximations $\sqrt{0} = 0$, $\sqrt{1} = 1$, $\sqrt{2} \approx 1.414$, $\sqrt{3} \approx 1.732$, $\sqrt{4} = 2$ are used for the function values.

Step 2. Formula used (Trapezoidal Rule, $n = 4$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$T_4 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)\Big].$$

Step 3. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{4-0}{4} = 1.$$

Step 4. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 0 \qquad x_1 = 1 \qquad x_2 = 2 \qquad x_3 = 3 \qquad x_4 = 4$$

Step 5. Function values at the grid points:

$$f(x_0) = \sqrt{0} = 0 \qquad f(x_1) = \sqrt{1} = 1$$
$$f(x_2) = \sqrt{2} = 1.414 \qquad f(x_3) = \sqrt{3} = 1.732$$
$$f(x_4) = \sqrt{4} = 2$$

Step 6. Insert the weights $(1, 2, 2, \ldots, 2, 1)$ and the function values:

$$T_4 = \frac{1}{2}\Big[\,0 + 2(1) + 2(1.414) + 2(1.732) + 2\,\Big]$$

Step 7. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \ 0 + 2 + 2.828 + 3.464 + 2 \ = \ 10.292$$

Step 8. Multiply by the front factor:

$$T_4 = 0.5 \times 10.292 = 5.146.$$

Step 9. Exact value for comparison: $\displaystyle\int_0^4 \sqrt{x}\,dx \ = \ \Big[\frac{2}{3}x^{3/2}\Big]_0^4 \ = \ \frac{2}{3}\cdot 8 \ = \ \frac{16}{3} \ \approx 5.333333.$

Step 10. Error:

$$T_4 - I = 5.146 - 5.33333 = -0.187333, \qquad |I - T_4| = 0.1873.$$

The difference is negative, so T_4 **underestimates** the exact value.

Step 11. This matches the concavity: $f''(x) = -\dfrac{1}{4}x^{-3/2} < 0$ for $x > 0$, so the graph is concave down, the chords lie below the graph, and T_4 underestimates.

Answer. $T_4 \approx 5.146$ (an underestimate of the exact value $\dfrac{16}{3} \approx 5.333$).

2. Approximate the same integral, $\displaystyle\int_0^4 \sqrt{x}\,dx$, using M_4 (midpoints $0.5, 1.5, 2.5, 3.5$; use $\sqrt{0.5} \approx 0.707,\ \sqrt{1.5} \approx 1.225,\ \sqrt{2.5} \approx 1.581,\ \sqrt{3.5} \approx 1.871$).

SOLUTION

Step 1. Setup: $f(x) = \sqrt{x}$ on $[0, 4]$ with $n = 4$, $\Delta x = 1$; the midpoints are $0.5, 1.5, 2.5, 3.5$ and the given approximations $\sqrt{0.5} \approx 0.707$, $\sqrt{1.5} \approx 1.225$, $\sqrt{2.5} \approx 1.581$, $\sqrt{3.5} \approx 1.871$ are used.

Step 2. Formula used (Midpoint Rule, $n = 4$):

$$M_n = \Delta x \Big[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n) \Big].$$

For $n = 4$ this is

$$M_4 = \Delta x \Big[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) + f(\bar{x}_4) \Big].$$

Step 3. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{4-0}{4} = 1.$$

Step 4. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$:

$$\bar{x}_1 = \frac{0+1}{2} = 0.5 \qquad \bar{x}_2 = \frac{1+2}{2} = 1.5$$

$$\bar{x}_3 = \frac{2+3}{2} = 2.5 \qquad \bar{x}_4 = \frac{3+4}{2} = 3.5$$

Step 5. Function values at the midpoints:

$$f(\bar{x}_1) = \sqrt{0.5} = 0.707 \qquad f(\bar{x}_2) = \sqrt{1.5} = 1.225$$

$$f(\bar{x}_3) = \sqrt{2.5} = 1.581 \qquad f(\bar{x}_4) = \sqrt{3.5} = 1.871$$

Step 6. Add the 4 midpoint values (all weights are 1):

$$\text{bracket} = \ 0.707 + 1.225 + 1.581 + 1.871 \ = \ 5.384$$

Step 7. Multiply by Δx:

$$M_4 = 1 \times 5.384 = 5.384.$$

Step 8. Error:

$$M_4 - I = 5.384 - 5.33333 = 0.0506667, \qquad |I - M_4| = 0.05067.$$

The difference is positive, so M_4 **overestimates** the exact value.

Step 9. This matches the concavity: f is concave down ($f''(x) = -\dfrac{1}{4}x^{-3/2} < 0$), the tangent line at each midpoint lies above the graph, so each midpoint rectangle has more area than the region under the curve and M_4 overestimates. (Compare Problem 1: T_4 underestimates while M_4 overestimates; here $|I - T_4| = 0.1873$ and $|I - M_4| = 0.0507$.)

Answer. $M_4 \approx 5.384$ (an overestimate of $\dfrac{16}{3} \approx 5.333$).

3. Approximate $\displaystyle\int_0^{\pi/2} \cos x\,dx$ using T_4 and M_4 (so $\Delta x = \dfrac{\pi}{8}$). Compare both with the exact value, and say whether each rule over- or under-estimates and why.

SOLUTION

Step 1. Trapezoidal Rule T_4.

Step 2. Formula used (Trapezoidal Rule, $n = 4$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$T_4 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)\Big].$$

Step 3. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{\frac{\pi}{2} - 0}{4} = \frac{\pi}{8} \approx 0.392699.$$

Step 4. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$x_0 = 0$ $\quad x_1 = \dfrac{\pi}{8} \approx 0.392699$ $\quad x_2 = \dfrac{\pi}{4} \approx 0.785398$ $\quad x_3 = \dfrac{3\pi}{8} \approx 1.1781$ $\quad x_4 = \dfrac{\pi}{2} \approx 1.5708$

Step 5. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = \cos(0) = 1 \qquad f(x_1) = \cos\left(\frac{\pi}{8}\right) = 0.92388$$

$$f(x_2) = \cos\left(\frac{\pi}{4}\right) = 0.707107 \qquad f(x_3) = \cos\left(\frac{3\pi}{8}\right) = 0.382683$$

$$f(x_4) = \cos\left(\frac{\pi}{2}\right) = 0$$

Step 6. Insert the weights $(1, 2, 2, \ldots, 2, 1)$ and the function values:

$$T_4 = \frac{\pi}{16}\Big[1 + 2(0.92388) + 2(0.707107) + 2(0.382683) + 0\Big]$$

Step 7. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = 1 + 1.84776 + 1.414214 + 0.765366 + 0 = 5.02734$$

Step 8. Multiply by the front factor:

$$T_4 = \frac{\pi}{16}\,(\approx 0.19635) \times 5.02734 = 0.987116.$$

Step 9. Midpoint Rule M_4.

Step 10. Formula used (Midpoint Rule, $n = 4$):

$$M_n = \Delta x\,\Big[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\Big].$$

For $n = 4$ this is

$$M_4 = \Delta x\Big[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) + f(\bar{x}_4)\Big].$$

Step 11. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{\frac{\pi}{2} - 0}{4} = \frac{\pi}{8} \approx 0.392699.$$

Step 12. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$:

$$\bar{x}_1 = \frac{0 + \frac{\pi}{8}}{2} = \frac{\pi}{16} \approx 0.19635$$
$$\bar{x}_2 = \frac{\frac{\pi}{8} + \frac{\pi}{4}}{2} = \frac{3\pi}{16} \approx 0.589049$$
$$\bar{x}_3 = \frac{\frac{\pi}{4} + \frac{3\pi}{8}}{2} = \frac{5\pi}{16} \approx 0.981748$$
$$\bar{x}_4 = \frac{\frac{3\pi}{8} + \frac{\pi}{2}}{2} = \frac{7\pi}{16} \approx 1.37445$$

Step 13. Function values at the midpoints (rounded to 6 significant figures where necessary):

$$f(\bar{x}_1) = \cos\left(\frac{\pi}{16}\right) = 0.980785 \qquad f(\bar{x}_2) = \cos\left(\frac{3\pi}{16}\right) = 0.83147$$
$$f(\bar{x}_3) = \cos\left(\frac{5\pi}{16}\right) = 0.55557 \qquad f(\bar{x}_4) = \cos\left(\frac{7\pi}{16}\right) = 0.19509$$

Step 14. Add the 4 midpoint values (all weights are 1):

$$\text{bracket} = 0.980785 + 0.83147 + 0.55557 + 0.19509 = 2.562915$$

Step 15. Multiply by Δx:

$$M_4 = \frac{\pi}{8}\,(\approx 0.392699) \times 2.562915 = 1.00645.$$

Step 16. Exact value: $\displaystyle\int_0^{\pi/2} \cos x\,dx = \Big[\sin x\Big]_0^{\pi/2} = \sin\frac{\pi}{2} - \sin 0 = 1.$

Step 17. Errors: $T_4 - 1 = -0.01288$ (underestimate) and $M_4 - 1 = +0.006454$ (overestimate). Reason: $f''(x) = -\cos x \le 0$ on $\left[0, \frac{\pi}{2}\right]$, so the graph is concave down; trapezoids lie below the curve and midpoint rectangles (tangent-line trapezoids) lie above it.

Answer. $T_4 \approx 0.98712$ (under), $M_4 \approx 1.0065$ (over); exact value 1.

4. The integral $\displaystyle\int_0^1 e^{-x^2}\,dx$ has no elementary antiderivative. Approximate it using T_4, M_4 and S_4, and compare each with the value 0.746824 obtained by computer.

SOLUTION

Step 1. The antiderivative of e^{-x^2} is not an elementary function, so the integral is approximated; use $f(x) = e^{-x^2}$, $a = 0$, $b = 1$, $n = 4$.

Step 2. Trapezoidal Rule T_4.

Step 3. Formula used (Trapezoidal Rule, $n = 4$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$T_4 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)\Big].$$

Step 4. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{4} = 0.25.$$

Step 5. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 0 \qquad x_1 = 0.25 \qquad x_2 = 0.5 \qquad x_3 = 0.75 \qquad x_4 = 1$$

Step 6. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = e^{-(0)^2} = 1 \qquad f(x_1) = e^{-(0.25)^2} = 0.939413$$
$$f(x_2) = e^{-(0.5)^2} = 0.778801 \qquad f(x_3) = e^{-(0.75)^2} = 0.569783$$
$$f(x_4) = e^{-(1)^2} = 0.367879$$

Step 7. Insert the weights $(1, 2, 2, \ldots, 2, 1)$ and the function values:

$$T_4 = \frac{0.25}{2}\Big[\,1 + 2(0.939413) + 2(0.778801) + 2(0.569783) + 0.367879\,\Big]$$

Step 8. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = 1 + 1.878826 + 1.557602 + 1.139566 + 0.367879 = 5.943873$$

Step 9. Multiply by the front factor:

$$T_4 = 0.125 \times 5.943873 = 0.742984.$$

Step 10. Midpoint Rule M_4.

Step 11. Formula used (Midpoint Rule, $n = 4$):

$$M_n = \Delta x\,\Big[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\Big].$$

For $n = 4$ this is

$$M_4 = \Delta x\Big[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) + f(\bar{x}_4)\Big].$$

Step 12. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{4} = 0.25.$$

Step 13. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$:

$$\bar{x}_1 = \frac{0+0.25}{2} = 0.125 \qquad \bar{x}_2 = \frac{0.25+0.5}{2} = 0.375$$
$$\bar{x}_3 = \frac{0.5+0.75}{2} = 0.625 \qquad \bar{x}_4 = \frac{0.75+1}{2} = 0.875$$

Step 14. Function values at the midpoints (rounded to 6 significant figures where necessary):

$$f(\bar{x}_1) = e^{-(0.125)^2} = 0.984496 \qquad f(\bar{x}_2) = e^{-(0.375)^2} = 0.868815$$
$$f(\bar{x}_3) = e^{-(0.625)^2} = 0.676634 \qquad f(\bar{x}_4) = e^{-(0.875)^2} = 0.465043$$

Step 15. Add the 4 midpoint values (all weights are 1):

$$\text{bracket} = 0.984496 + 0.868815 + 0.676634 + 0.465043 = 2.994988$$

Step 16. Multiply by Δx:

$$M_4 = 0.25 \times 2.994988 = 0.748747.$$

Step 17. Simpson's Rule S_4.

Step 18. Formula used (Simpson's Rule, $n = 4$):

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$S_4 = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)\Big].$$

Step 19. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{4} = 0.25.$$

Step 20. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 0 \qquad x_1 = 0.25 \qquad x_2 = 0.5 \qquad x_3 = 0.75 \qquad x_4 = 1$$

Step 21. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = e^{-(0)^2} = 1 \qquad f(x_1) = e^{-(0.25)^2} = 0.939413$$
$$f(x_2) = e^{-(0.5)^2} = 0.778801 \qquad f(x_3) = e^{-(0.75)^2} = 0.569783$$
$$f(x_4) = e^{-(1)^2} = 0.367879$$

Step 22. Insert the weights $(1, 4, 2, 4, \ldots, 2, 4, 1)$ and the function values:

$$S_4 = \frac{0.25}{3}\Big[\,1 + 4(0.939413) + 2(0.778801) + 4(0.569783) + 0.367879\,\Big]$$

Step 23. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = 1 + 3.757652 + 1.557602 + 2.279132 + 0.367879 = 8.962265$$

Step 24. Multiply by the front factor:

$$S_4 = \frac{0.25}{3}\,(\approx 0.0833333) \times 8.962265 = 0.746855.$$

Step 25. Compare with the computer value 0.746824 (6 decimal places):

$$|0.746824 - T_4| = 0.00384, \qquad |0.746824 - M_4| = 0.00192, \qquad |0.746824 - S_4| = 0.0000314.$$

Step 26. Ranking from best to worst: S_4 (error about 0.000031), then M_4 (0.0019), then T_4 (0.0038). Note $T_4 < I < M_4$ here; the concavity argument does not apply because $f''(x) = (4x^2 - 2)e^{-x^2}$ changes sign on $[0, 1]$ (at $x = \dfrac{1}{\sqrt{2}} \approx 0.7071$).

Answer. $T_4 \approx 0.74298$, $M_4 \approx 0.74875$, $S_4 \approx 0.74686$; Simpson is the most accurate.

5. Approximate the area of the half-disc $\displaystyle\int_{-1}^{1} \sqrt{1-x^2}\,dx$ using T_4 and M_4. Compare with the exact area $\dfrac{\pi}{2}$, explain the signs of the errors using f'', and explain why the error-bound theorem cannot be used for this integral.

SOLUTION

Step 1. The integral is the area of the upper half of the unit disc, so its exact value is $\dfrac{1}{2}\pi(1)^2 = \dfrac{\pi}{2} \approx 1.570796$. Use $f(x) = \sqrt{1-x^2}$ on $[-1, 1]$, $n = 4$.

Step 2. Trapezoidal Rule T_4.

Step 3. Formula used (Trapezoidal Rule, $n = 4$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$T_4 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)\Big].$$

Step 4. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{1-(-1)}{4} = 0.5.$$

Step 5. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = -1 \qquad x_1 = -0.5 \qquad x_2 = 0 \qquad x_3 = 0.5 \qquad x_4 = 1$$

Step 6. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = \sqrt{1-(-1)^2} = 0 \qquad f(x_1) = \sqrt{1-(-0.5)^2} = 0.866025$$
$$f(x_2) = \sqrt{1-(0)^2} = 1 \qquad f(x_3) = \sqrt{1-(0.5)^2} = 0.866025$$
$$f(x_4) = \sqrt{1-(1)^2} = 0$$

Step 7. Insert the weights $(1, 2, 2, \ldots, 2, 1)$ and the function values:

$$T_4 = \frac{0.5}{2}\Big[\,0 + 2(0.866025) + 2(1) + 2(0.866025) + 0\,\Big]$$

Step 8. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; 0 + 1.73205 + 2 + 1.73205 + 0 \; = \; 5.4641$$

Step 9. Multiply by the front factor:

$$T_4 = 0.25 \times 5.4641 = 1.36603.$$

Step 10. Midpoint Rule M_4.

Step 11. Formula used (Midpoint Rule, $n = 4$):

$$M_n = \Delta x\,\Big[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\Big].$$

For $n = 4$ this is

$$M_4 = \Delta x\Big[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) + f(\bar{x}_4)\Big].$$

Step 12. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{1-(-1)}{4} = 0.5.$$

Step 13. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1}+x_i}{2}$:

$$\bar{x}_1 = \frac{(-1)+(-0.5)}{2} = -0.75 \qquad \bar{x}_2 = \frac{(-0.5)+0}{2} = -0.25$$
$$\bar{x}_3 = \frac{0+0.5}{2} = 0.25 \qquad \bar{x}_4 = \frac{0.5+1}{2} = 0.75$$

Step 14. Function values at the midpoints (rounded to 6 significant figures where necessary):

$$f(\bar{x}_1) = \sqrt{1-(-0.75)^2} = 0.661438 \qquad f(\bar{x}_2) = \sqrt{1-(-0.25)^2} = 0.968246$$
$$f(\bar{x}_3) = \sqrt{1-(0.25)^2} = 0.968246 \qquad f(\bar{x}_4) = \sqrt{1-(0.75)^2} = 0.661438$$

Step 15. Add the 4 midpoint values (all weights are 1):

$$\text{bracket} = 0.661438 + 0.968246 + 0.968246 + 0.661438 = 3.259368$$

Step 16. Multiply by Δx:

$$M_4 = 0.5 \times 3.259368 = 1.62968.$$

Step 17. Error:

$$T_4 - I = 1.36603 - 1.5708 = -0.204771, \qquad |I - T_4| = 0.2048.$$

The difference is negative, so T_4 **underestimates** the exact value.

Step 18. Error:

$$M_4 - I = 1.62968 - 1.5708 = 0.0588877, \qquad |I - M_4| = 0.05889.$$

The difference is positive, so M_4 **overestimates** the exact value.

Step 19. Concavity: $f'(x) = -\dfrac{x}{\sqrt{1-x^2}}$ and $f''(x) = -\dfrac{1}{(1-x^2)^{3/2}} < 0$ on $(-1,1)$, so the graph is concave down: trapezoids underestimate and midpoint rectangles overestimate, as found.

Step 20. Error bound: $f''(x) = -(1-x^2)^{-3/2} \to -\infty$ as $x \to \pm 1$, so no finite K with $|f''(x)| \le K$ on $[-1,1]$ exists. The error-bound theorem therefore cannot be applied, and the errors are larger than the bounds would suggest for a smooth function (for instance $|I - T_4|$ is about 3.5 times $|I - M_4|$, not 2 times).

Answer. $T_4 \approx 1.366$ (under), $M_4 \approx 1.6297$ (over); exact $\dfrac{\pi}{2} \approx 1.5708$; f'' is unbounded near ± 1.

Simpson's Rule

6. Approximate $\displaystyle\int_0^2 (x^3+1)\,dx$ using S_4 (grid points $0, 0.5, 1, 1.5, 2$), and compare to the exact value. Explain why the two must match exactly.

SOLUTION

Step 1. Formula used (Simpson's Rule, $n = 4$):

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$S_4 = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)\Big].$$

Step 2. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{4} = 0.5.$$

Step 3. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 0 \quad x_1 = 0.5 \quad x_2 = 1 \quad x_3 = 1.5 \quad x_4 = 2$$

Step 4. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = (0)^3 + 1 = 1 \qquad f(x_1) = (0.5)^3 + 1 = 1.125$$
$$f(x_2) = (1)^3 + 1 = 2 \qquad f(x_3) = (1.5)^3 + 1 = 4.375$$
$$f(x_4) = (2)^3 + 1 = 9$$

Step 5. Insert the weights $(1, 4, 2, 4, \ldots, 2, 4, 1)$ and the function values:

$$S_4 = \frac{0.5}{3}\Big[\,1 + 4(1.125) + 2(2) + 4(4.375) + 9\,\Big]$$

Step 6. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; 1 + 4.5 + 4 + 17.5 + 9 \; = \; 36$$

Step 7. Multiply by the front factor:

$$S_4 = \frac{0.5}{3}\,(\approx 0.166667) \times 36 = 6.$$

Step 8. Exact value: $\displaystyle\int_0^2 (x^3+1)\,dx = \Big[\frac{x^4}{4} + x\Big]_0^2 = \Big(\frac{16}{4} + 2\Big) - (0+0) = 4 + 2 = 6.$

Step 9. Comparison: $S_4 = 6$ equals the exact value 6, so the error is exactly 0.

Step 10. Why the two must match: Simpson's Rule integrates every polynomial of degree ≤ 3 exactly, and $f(x) = x^3 + 1$ has degree 3. Equivalently, $f^{(4)}(x) = 0$ for all x (differentiate: $3x^2$, $6x$, 6, 0), so the error bound $\dfrac{M(b-a)^5}{180n^4}$ has $M = 0$ and the error is 0.

Answer. $S_4 = 6$, equal to the exact value 6 (Simpson is exact for polynomials of degree ≤ 3).

7. Approximate $\displaystyle\int_0^{\pi} \sin x\,dx$ using S_4 (grid points $0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$; use $\sin 0 = 0,\ \sin\frac{\pi}{4} \approx 0.707,\ \sin\frac{\pi}{2} = 1,\ \sin\frac{3\pi}{4} \approx 0.707,\ \sin\pi = 0$), and compare to the exact value 2.

SOLUTION

Step 1. Setup: $f(x) = \sin x$ on $[0, \pi]$, $n = 4$, with the given values $\sin 0 = 0$, $\sin\frac{\pi}{4} \approx 0.707$, $\sin\frac{\pi}{2} = 1$, $\sin\frac{3\pi}{4} \approx 0.707$, $\sin\pi = 0$.

Step 2. Formula used (Simpson's Rule, $n = 4$):

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$S_4 = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)\Big].$$

Step 3. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{\pi - 0}{4} = \frac{\pi}{4} \approx 0.785398.$$

Step 4. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$x_0 = 0$ $\quad x_1 = \frac{\pi}{4} \approx 0.785398$ $\quad x_2 = \frac{\pi}{2} \approx 1.5708$ $\quad x_3 = \frac{3\pi}{4} \approx 2.35619$ $\quad x_4 = \pi \approx 3.14159$

Step 5. Function values at the grid points:

$$f(x_0) = \sin(0) = 0 \qquad f(x_1) = \sin\left(\frac{\pi}{4}\right) = 0.707$$

$$f(x_2) = \sin\left(\frac{\pi}{2}\right) = 1 \qquad f(x_3) = \sin\left(\frac{3\pi}{4}\right) = 0.707$$

$$f(x_4) = \sin(\pi) = 0$$

Step 6. Insert the weights $(1, 4, 2, 4, \ldots, 2, 4, 1)$ and the function values:

$$S_4 = \frac{\pi}{12}\Big[\,0 + 4(0.707) + 2(1) + 4(0.707) + 0\,\Big]$$

Step 7. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \ 0 + 2.828 + 2 + 2.828 + 0 \ = \ 7.656$$

Step 8. Multiply by the front factor:

$$S_4 = \frac{\pi}{12}\,(\approx 0.261799) \times 7.656 = 2.00434.$$

Step 9. Exact value: $\displaystyle\int_0^{\pi} \sin x\,dx = \Big[-\cos x\Big]_0^{\pi} = -\cos\pi + \cos 0 = 1 + 1 = 2$. So the error is $|2 - 2.0043| = 0.00434$, i.e. about 0.22% of the exact value.

Step 10. Why it is close but not exact: $\sin x$ is not a polynomial of degree ≤ 3 ($f^{(4)}(x) = \sin x \neq 0$). The bound with $M = 1$ is $\dfrac{1 \cdot \pi^5}{180 \cdot 4^4} = 0.006641$, and the error 0.00434 lies below it.

Answer. $S_4 \approx 2.004$ against the exact value 2 (error about 0.004).

8. Approximate $\displaystyle\int_1^2 \frac{1}{x}\,dx$ using Simpson's Rule S_4 and compare with the exact value $\ln 2$.

SOLUTION

Step 1. Formula used (Simpson's Rule, $n = 4$):

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$S_4 = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)\Big].$$

Step 2. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{2-1}{4} = 0.25.$$

Step 3. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 1 \qquad x_1 = 1.25 \qquad x_2 = 1.5 \qquad x_3 = 1.75 \qquad x_4 = 2$$

Step 4. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = \frac{1}{1} = 1 \qquad f(x_1) = \frac{1}{1.25} = 0.8$$

$$f(x_2) = \frac{1}{1.5} = 0.666667 \qquad f(x_3) = \frac{1}{1.75} = 0.571429$$

$$f(x_4) = \frac{1}{2} = 0.5$$

Step 5. Insert the weights $(1, 4, 2, 4, \ldots, 2, 4, 1)$ and the function values:

$$S_4 = \frac{0.25}{3}\Big[\,1 + 4(0.8) + 2(0.666667) + 4(0.571429) + 0.5\,\Big]$$

Step 6. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = 1 + 3.2 + 1.333334 + 2.285716 + 0.5 = 8.31905$$

Step 7. Multiply by the front factor:

$$S_4 = \frac{0.25}{3}\,(\approx 0.0833333) \times 8.31905 = 0.693254.$$

Step 8. Exact value: $\displaystyle\int_1^2 \frac{dx}{x} = \Big[\ln x\Big]_1^2 = \ln 2 - \ln 1 = \ln 2 \approx 0.693147.$

Step 9. Error: $S_4 - \ln 2 = 0.693254 - 0.693147 = 0.000107$, so $|\ln 2 - S_4| = 0.000107$ (about 0.015% of $\ln 2$), a slight overestimate. Compare with the Trapezoidal Rule in Example 4: even with $n = 4$ the Simpson error is far smaller.

Answer. $S_4 \approx 0.693254$ against $\ln 2 \approx 0.693147$ (error about 0.00011).

9. Use Simpson's Rule with $n = 2$ to approximate $\displaystyle\int_{-1}^{3} (x^3 - 2x^2 + 4x - 1)\,dx$. Compute the exact value and explain the result.

SOLUTION

Step 1. Formula used (Simpson's Rule, $n = 2$):

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big].$$

For $n = 2$ this is

$$S_2 = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + f(x_2)\Big].$$

Step 2. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{3-(-1)}{2} = 2.$$

Step 3. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 2$:

$$x_0 = -1 \qquad x_1 = 1 \qquad x_2 = 3$$

Step 4. Function values at the grid points (rounded to 6 significant figures where necessary):

$f(x_0) = (-1)^3 - 2(-1)^2 + 4(-1) - 1 = -8$ $\qquad$ $f(x_1) = (1)^3 - 2(1)^2 + 4(1) - 1 = 2$

$f(x_2) = (3)^3 - 2(3)^2 + 4(3) - 1 = 20$

Step 5. Insert the weights $(1, 4, 2, 4, \ldots, 2, 4, 1)$ and the function values:

$$S_2 = \frac{2}{3}\Big[\,-8 + 4(2) + 20\,\Big]$$

Step 6. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; -8 + 8 + 20 \; = \; 20$$

Step 7. Multiply by the front factor:

$$S_2 = \frac{2}{3}\,(\approx 0.666667) \times 20 = 13.3333.$$

Step 8. Exact value, term by term:

$$\int_{-1}^{3}(x^3 - 2x^2 + 4x - 1)\,dx = \left[\frac{x^4}{4} - \frac{2x^3}{3} + 2x^2 - x\right]_{-1}^{3}.$$

Step 9. Upper limit $x = 3$: $\dfrac{81}{4} - \dfrac{2 \cdot 27}{3} + 2 \cdot 9 - 3 = 20.25 - 18 + 18 - 3 = 17.25.$

Lower limit $x = -1$: $\dfrac{1}{4} - \dfrac{2(-1)}{3} + 2(1) - (-1) = 0.25 + 0.666667 + 2 + 1 = 3.916667.$

Subtract: $17.25 - 3.916667 = 13.333333 = \dfrac{40}{3}.$

Step 10. Comparison: $S_2 = \dfrac{40}{3}$ equals the exact value, with only $n = 2$ subintervals (three function values). Reason: f is a cubic, $f^{(4)}(x) = 0$, so the Simpson error bound is 0 for every even n.

Answer. $S_2 = \dfrac{40}{3} \approx 13.3333$, exactly equal to the integral.

10. Let $f(x) = x^4$ on $[0, 2]$. (a) Compute S_2 and its exact error. (b) Compute S_4 and its exact error. (c) Compare each error with the Simpson error bound and find the ratio of the two errors.

SOLUTION

Step 1. Exact value: $\displaystyle\int_0^2 x^4\,dx = \left[\frac{x^5}{5}\right]_0^2 = \frac{32}{5} = 6.4.$

Step 2. (a) Simpson with $n = 2$.

Step 3. Formula used (Simpson's Rule, $n = 2$):

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big].$$

For $n = 2$ this is

$$S_2 = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + f(x_2)\Big].$$

Step 4. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{2} = 1.$$

Step 5. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 2$:

$$x_0 = 0 \qquad x_1 = 1 \qquad x_2 = 2$$

Step 6. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = (0)^4 = 0 \qquad f(x_1) = (1)^4 = 1$$
$$f(x_2) = (2)^4 = 16$$

Step 7. Insert the weights $(1, 4, 2, 4, \ldots, 2, 4, 1)$ and the function values:

$$S_2 = \frac{1}{3}\Big[\,0 + 4(1) + 16\,\Big]$$

Step 8. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; 0 + 4 + 16 \; = \; 20$$

Step 9. Multiply by the front factor:

$$S_2 = \frac{1}{3}\,(\approx 0.333333) \times 20 = 6.66667.$$

Step 10. Error: $S_2 - \dfrac{32}{5} = \dfrac{20}{3} - \dfrac{32}{5} = \dfrac{100-96}{15} = \dfrac{4}{15} = 0.266667.$ The rule is **not** exact here because x^4 has degree $4 > 3$.

Step 11. Error bound: $f^{(4)}(x) = 24$ is constant, so $M = 24$ and

$$E_S \le \frac{24 \cdot 2^5}{180 \cdot 2^4} = \frac{768}{2880} = \frac{4}{15} = 0.266667.$$

The actual error **equals** the bound, so the bound cannot be improved in general.

Step 12. (b) Simpson with $n = 4$.

Step 13. Formula used (Simpson's Rule, $n = 4$):

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$S_4 = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)\Big].$$

Step 14. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{2-0}{4} = 0.5.$$

Step 15. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 0 \qquad x_1 = 0.5 \qquad x_2 = 1 \qquad x_3 = 1.5 \qquad x_4 = 2$$

Step 16. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = (0)^4 = 0 \qquad f(x_1) = (0.5)^4 = 0.0625$$
$$f(x_2) = (1)^4 = 1 \qquad f(x_3) = (1.5)^4 = 5.0625$$
$$f(x_4) = (2)^4 = 16$$

Step 17. Insert the weights $(1, 4, 2, 4, \ldots, 2, 4, 1)$ and the function values:

$$S_4 = \frac{0.5}{3}\Big[\,0 + 4(0.0625) + 2(1) + 4(5.0625) + 16\,\Big]$$

Step 18. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; 0 + 0.25 + 2 + 20.25 + 16 \; = \; 38.5$$

Step 19. Multiply by the front factor:

$$S_4 = \frac{0.5}{3}\,(\approx 0.166667) \times 38.5 = 6.41667.$$

Step 20. Error: $S_4 - 6.4 = 6.41667 - 6.4 = 0.01667$ (as a fraction, $\frac{1}{60}$). The bound is $\dfrac{24 \cdot 2^5}{180 \cdot 4^4} = \dfrac{768}{46080} = \dfrac{1}{60} = 0.016667$, again equal to the actual error.

Step 21. Ratio of the errors: $\dfrac{4/15}{1/60} = \dfrac{4 \cdot 60}{15} = 16$, exactly the factor $2^4 = 16$ predicted by the n^4 in the denominator when n doubles from 2 to 4.

Answer. $S_2 = \dfrac{20}{3}$ (error $\dfrac{4}{15}$), $S_4 = \dfrac{77}{12}$ (error $\dfrac{1}{60}$); both errors equal their bounds, and the ratio is 16.

Numerical integration from tables of data

11. A car accelerates from rest and its speed $v(t)$ (m/s) is measured every 2 seconds:

t (s)	0	2	4	6	8	10	12
$v(t)$	0	6.6	11.0	14.0	16.0	17.3	18.2

Estimate the distance travelled in the first 12 seconds, $\displaystyle\int_0^{12} v(t)\,dt$, using (a) T_6, (b) S_6, (c) M_3 (with $n = 3$ subintervals of width 4 s).

SOLUTION

Step 1. The distance travelled is $s = \displaystyle\int_0^{12} v(t)\,dt$ (metres, with t in seconds and v in metres per second). The speeds are given at $t = 0, 2, \dots, 12$, spacing 2 s, so $n = 6$ subintervals.

Step 2. (a) Trapezoidal Rule T_6.

Step 3. Formula used (Trapezoidal Rule, $n = 6$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 6$ this is

$$T_6 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_5) + f(x_6)\Big].$$

Step 4. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{12-0}{6} = 2.$$

Step 5. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \dots, 6$:

$$x_0 = 0 \quad x_1 = 2 \quad x_2 = 4 \quad x_3 = 6$$
$$x_4 = 8 \quad x_5 = 10 \quad x_6 = 12$$

Step 6. Function values at the grid points:

$$f(x_0) = v(0) = 0 \qquad f(x_1) = v(2) = 6.6$$
$$f(x_2) = v(4) = 11 \qquad f(x_3) = v(6) = 14$$
$$f(x_4) = v(8) = 16 \qquad f(x_5) = v(10) = 17.3$$
$$f(x_6) = v(12) = 18.2$$

Step 7. Insert the weights $(1, 2, 2, \dots, 2, 1)$ and the function values:

$$T_6 = \frac{2}{2}\Big[\,0 + 2(6.6) + 2(11) + 2(14) + 2(16) + 2(17.3) + 18.2\,\Big]$$

Step 8. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = 0 + 13.2 + 22 + 28 + 32 + 34.6 + 18.2 = 148$$

Step 9. Multiply by the front factor:

$$T_6 = 1 \times 148 = 148.$$

Step 10. (b) Simpson's Rule S_6 ($n = 6$ is even).

Step 11. Formula used (Simpson's Rule, $n = 6$):

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big].$$

For $n = 6$ this is

$$S_6 = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + 4f(x_5) + f(x_6)\Big].$$

Step 12. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{12-0}{6} = 2.$$

Step 13. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 6$:

$$x_0 = 0 \qquad x_1 = 2 \qquad x_2 = 4 \qquad x_3 = 6$$
$$x_4 = 8 \qquad x_5 = 10 \qquad x_6 = 12$$

Step 14. Function values at the grid points:

$$f(x_0) = v(0) = 0 \qquad f(x_1) = v(2) = 6.6$$
$$f(x_2) = v(4) = 11 \qquad f(x_3) = v(6) = 14$$
$$f(x_4) = v(8) = 16 \qquad f(x_5) = v(10) = 17.3$$
$$f(x_6) = v(12) = 18.2$$

Step 15. Insert the weights $(1, 4, 2, 4, \ldots, 2, 4, 1)$ and the function values:

$$S_6 = \frac{2}{3}\Big[\,0 + 4(6.6) + 2(11) + 4(14) + 2(16) + \;4(17.3) + 18.2\,\Big]$$

Step 16. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \;0 + 26.4 + 22 + 56 + 32 + 69.2 + \;18.2 \;=\; 223.8$$

Step 17. Multiply by the front factor:

$$S_6 = \frac{2}{3}\,(\approx 0.666667) \times 223.8 = 149.2.$$

Step 18. **(c) Midpoint Rule.** Use subintervals of width 4 ($n = 3$: $[0, 4], [4, 8], [8, 12]$), whose midpoints $2, 6, 10$ are in the table.

Step 19. Formula used (Midpoint Rule, $n = 3$):

$$M_n = \Delta x\,\Big[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\Big].$$

For $n = 3$ this is

$$M_3 = \Delta x\Big[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3)\Big].$$

Step 20. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{12-0}{3} = 4.$$

Step 21. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$:

$$\bar{x}_1 = \frac{0+4}{2} = 2 \qquad \bar{x}_2 = \frac{4+8}{2} = 6$$
$$\bar{x}_3 = \frac{8+12}{2} = 10$$

Step 22. Function values at the midpoints:

$$f(\bar{x}_1) = v(2) = 6.6 \qquad f(\bar{x}_2) = v(6) = 14$$
$$f(\bar{x}_3) = v(10) = 17.3$$

Step 23. Add the 3 midpoint values (all weights are 1):

$$\text{bracket} = 6.6 + 14 + 17.3 = 37.9$$

Step 24. Multiply by Δx:

$$M_3 = 4 \times 37.9 = 151.6.$$

Step 25. Concavity of the data: the successive increases of v are $6.6,\ 4.4,\ 3.0,\ 2.0,\ 1.3,\ 0.9$, which are decreasing, so the graph of v is concave down. For a concave-down function T_n underestimates and M_n overestimates, so the distance should lie between T_6 and M_3.

Step 26. Summary: $T_6 = 148$ m, $S_6 = 149.2$ m, $M_3 = 151.6$ m, consistent with $T_6 < S_6 < M_3$; the best single estimate is $S_6 \approx 149.2$ m.

Answer. $T_6 = 148$ m, $S_6 = 149.2$ m, $M_3 = 151.6$ m.

12. The depth $d(x)$ (metres) of a river was measured every 10 m across its width:

x (m)	0	10	20	30	40	50	60	70	80
$d(x)$	0	2.1	3.4	4.2	4.6	4.1	3.3	1.8	0

Using the table: (a) estimate the cross-sectional area $\displaystyle\int_0^{80} d(x)\,dx$ using T_8; (b) estimate it using S_8; (c) estimate it using M_4 (subintervals of width 20 m); (d) if the water flows at an average speed of 0.8 m/s, estimate the discharge (area $\times$ speed) using your Simpson area.

SOLUTION

Step 1. The cross-sectional area is $A = \displaystyle\int_0^{80} d(x)\,dx$ (square metres), with the depth d measured every 10 m, so $n = 8$ subintervals of width 10.

Step 2. (a) Trapezoidal Rule T_8.

Step 3. Formula used (Trapezoidal Rule, $n = 8$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 8$ this is

$$T_8 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_7) + f(x_8)\Big].$$

Step 4. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{80-0}{8} = 10.$$

Step 5. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 8$:

$$x_0 = 0 \quad x_1 = 10 \quad x_2 = 20 \quad x_3 = 30$$
$$x_4 = 40 \quad x_5 = 50 \quad x_6 = 60 \quad x_7 = 70$$
$$x_8 = 80$$

Step 6. Function values at the grid points:

$$f(x_0) = d(0) = 0 \qquad f(x_1) = d(10) = 2.1$$
$$f(x_2) = d(20) = 3.4 \qquad f(x_3) = d(30) = 4.2$$
$$f(x_4) = d(40) = 4.6 \qquad f(x_5) = d(50) = 4.1$$
$$f(x_6) = d(60) = 3.3 \qquad f(x_7) = d(70) = 1.8$$
$$f(x_8) = d(80) = 0$$

Step 7. Insert the weights $(1, 2, 2, \dots, 2, 1)$ and the function values:

$$T_8 = \frac{10}{2}\Big[\,0 + 2(2.1) + 2(3.4) + 2(4.2) + 2(4.6) + 2(4.1) + 2(3.3) + 2(1.8) + 0\,\Big]$$

Step 8. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = 0 + 4.2 + 6.8 + 8.4 + 9.2 + 8.2 + 6.6 + 3.6 + 0 = 47$$

Step 9. Multiply by the front factor:

$$T_8 = 5 \times 47 = 235.$$

Step 10. (b) Simpson's Rule S_8 ($n = 8$ is even).

Step 11. Formula used (Simpson's Rule, $n = 8$):

$$S_n = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)\Big].$$

For $n = 8$ this is

$$S_8 = \frac{\Delta x}{3}\Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 2f(x_6) + 4f(x_7) + f(x_8)\Big].$$

Step 12. Width of each subinterval:

$$\Delta x = \frac{b - a}{n} = \frac{80 - 0}{8} = 10.$$

Step 13. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \dots, 8$:

$$x_0 = 0 \quad x_1 = 10 \quad x_2 = 20 \quad x_3 = 30$$
$$x_4 = 40 \quad x_5 = 50 \quad x_6 = 60 \quad x_7 = 70$$
$$x_8 = 80$$

Step 14. Function values at the grid points:

$$f(x_0) = d(0) = 0 \qquad f(x_1) = d(10) = 2.1$$
$$f(x_2) = d(20) = 3.4 \qquad f(x_3) = d(30) = 4.2$$
$$f(x_4) = d(40) = 4.6 \qquad f(x_5) = d(50) = 4.1$$
$$f(x_6) = d(60) = 3.3 \qquad f(x_7) = d(70) = 1.8$$
$$f(x_8) = d(80) = 0$$

Step 15. Insert the weights $(1, 4, 2, 4, \dots, 2, 4, 1)$ and the function values:

$$S_8 = \frac{10}{3}\Big[\,0 + 4(2.1) + 2(3.4) + 4(4.2) + 2(4.6) + 4(4.1) + 2(3.3) + 4(1.8) + 0\,\Big]$$

Step 16. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = 0 + 8.4 + 6.8 + 16.8 + 9.2 + 16.4 + 6.6 + 7.2 + 0 = 71.4$$

Step 17. Multiply by the front factor:

$$S_8 = \frac{10}{3}\,(\approx 3.33333) \times 71.4 = 238.$$

Step 18. (c) Midpoint Rule. Use subintervals of width 20 ($n = 4$), whose midpoints $10, 30, 50, 70$ are in the table.

Step 19. Formula used (Midpoint Rule, $n = 4$):

$$M_n = \Delta x \left[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n) \right].$$

For $n = 4$ this is

$$M_4 = \Delta x \Big[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) + f(\bar{x}_4) \Big].$$

Step 20. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{80-0}{4} = 20.$$

Step 21. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$:

$$\bar{x}_1 = \frac{0+20}{2} = 10 \qquad \bar{x}_2 = \frac{20+40}{2} = 30$$
$$\bar{x}_3 = \frac{40+60}{2} = 50 \qquad \bar{x}_4 = \frac{60+80}{2} = 70$$

Step 22. Function values at the midpoints:

$$f(\bar{x}_1) = d(10) = 2.1 \qquad f(\bar{x}_2) = d(30) = 4.2$$
$$f(\bar{x}_3) = d(50) = 4.1 \qquad f(\bar{x}_4) = d(70) = 1.8$$

Step 23. Add the 4 midpoint values (all weights are 1):

$$\text{bracket} = 2.1 + 4.2 + 4.1 + 1.8 = 12.2$$

Step 24. Multiply by Δx:

$$M_4 = 20 \times 12.2 = 244.$$

Step 25. (d) Discharge. Take S_8 as the best estimate of the area, $A \approx 238$ m^2. Discharge = area $\times$ speed $= 238 \times 0.8 = 190.4$ m^3/s.

Answer. $T_8 = 235$ m^2, $S_8 = 238$ m^2, $M_4 = 244$ m^2; discharge ≈ 190.4 m^3/s.

Error bounds

13. Find a bound on the error in approximating $\displaystyle\int_0^1 e^x\,dx$ using T_8.

SOLUTION

Step 1. Identify: $f(x) = e^x$, $a = 0$, $b = 1$, $n = 8$, and $E_T \le \dfrac{K(b-a)^3}{12n^2}$ with $|f''(x)| \le K$ on $[0,1]$.

Step 2. Differentiate: $f'(x) = e^x$ and $f''(x) = e^x$.

Step 3. Maximise $|f''|$ on $[0,1]$: $e^x > 0$, so $|f''(x)| = e^x$. The exponential is increasing, so the maximum is at the right endpoint:

$$K = f''(1) = e \approx 2.718.$$

Step 4. Substitute $K = 2.718$, $b - a = 1$, $n = 8$:

$$E_T \le \frac{2.718 \cdot 1^3}{12 \cdot 8^2} = \frac{2.718}{12 \cdot 64} = \frac{2.718}{768}.$$

Step 5. Divide: $\dfrac{2.718}{768} = 0.003539$. So $\left|\displaystyle\int_0^1 e^x\,dx - T_8\right| \le 0.00354$.

Step 6. Reference check: $T_8 = 1.72052$ and $e - 1 = 1.718282$, so the actual error is $0.00224 \le 0.00354$ (the bound holds).

Answer. $E_T \le \dfrac{e}{12 \cdot 64} \approx 0.00354$.

14. Find a bound on the error in approximating $\displaystyle\int_1^3 \ln x\,dx$ using M_6.

SOLUTION

Step 1. Identify: $f(x) = \ln x$, $a = 1$, $b = 3$, $n = 6$, and $E_M \le \dfrac{K(b-a)^3}{24n^2}$ with $|f''(x)| \le K$ on $[1,3]$.

Step 2. Differentiate: $f'(x) = \dfrac{1}{x}$ and $f''(x) = -\dfrac{1}{x^2}$.

Step 3. Maximise $|f''|$ on $[1,3]$: $|f''(x)| = \dfrac{1}{x^2}$, which is decreasing for $x > 0$ (its derivative is $-\dfrac{2}{x^3} < 0$). The maximum is at $x = 1$:

$$K = \left|-\frac{1}{1^2}\right| = 1.$$

Step 4. Substitute $K = 1$, $b - a = 3 - 1 = 2$, $n = 6$:

$$E_M \le \frac{1 \cdot 2^3}{24 \cdot 6^2} = \frac{8}{24 \cdot 36} = \frac{8}{864}.$$

Step 5. Simplify: $\dfrac{8}{864} = \dfrac{1}{108} = 0.009259$. So $\left|\displaystyle\int_1^3 \ln x\,dx - M_6\right| \le 0.00926$.

Step 6. Reference check: $\displaystyle\int_1^3 \ln x\,dx = \Big[x\ln x - x\Big]_1^3 = 3\ln 3 - 2 = 1.29584$ and $M_6 = 1.2989$, so the actual error is $0.00306 \le 0.00926$.

Answer. $E_M \le \dfrac{1}{108} \approx 0.00926.$

15. Tasks: (a) find a bound on the error in approximating $\displaystyle\int_1^4 \sqrt{x}\,dx$ using T_6; (b) compute T_6 and the exact value, and check that the actual error respects your bound.

SOLUTION

Step 1. Part 1: the error bound. $f(x) = \sqrt{x} = x^{1/2}$, $a = 1$, $b = 4$, $n = 6$. Differentiate: $f'(x) = \dfrac{1}{2}x^{-1/2}$ and $f''(x) = -\dfrac{1}{4}x^{-3/2}$.

Step 2. Maximise $|f''|$ on $[1,4]$: $|f''(x)| = \dfrac{1}{4}x^{-3/2} = \dfrac{1}{4x^{3/2}}$, which decreases as x increases, so the maximum is at $x = 1$:

$$K = \frac{1}{4\cdot 1^{3/2}} = \frac{1}{4}.$$

Step 3. Substitute into $E_T \le \dfrac{K(b-a)^3}{12n^2}$ with $b - a = 3$:

$$E_T \le \frac{\frac{1}{4}\cdot 3^3}{12\cdot 6^2} = \frac{\frac{27}{4}}{432} = \frac{27}{1728} = \frac{1}{64} = 0.015625.$$

Step 4. Part 2: compute T_6 and check the bound.

Step 5. Formula used (Trapezoidal Rule, $n = 6$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 6$ this is

$$T_6 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + \cdots + 2f(x_5) + f(x_6)\Big].$$

Step 6. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{4-1}{6} = 0.5.$$

Step 7. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 6$:

$$x_0 = 1 \quad x_1 = 1.5 \quad x_2 = 2 \quad x_3 = 2.5$$
$$x_4 = 3 \quad x_5 = 3.5 \quad x_6 = 4$$

Step 8. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = \sqrt{1} = 1 \qquad f(x_1) = \sqrt{1.5} = 1.22474$$
$$f(x_2) = \sqrt{2} = 1.41421 \qquad f(x_3) = \sqrt{2.5} = 1.58114$$
$$f(x_4) = \sqrt{3} = 1.73205 \qquad f(x_5) = \sqrt{3.5} = 1.87083$$
$$f(x_6) = \sqrt{4} = 2$$

Step 9. Insert the weights $(1, 2, 2, \dots, 2, 1)$ and the function values:

$$T_6 = \frac{0.5}{2}\Big[\,1 + 2(1.22474) + 2(1.41421) + 2(1.58114) + 2(1.73205) + 2(1.87083) + 2\,\Big]$$

Step 10. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = 1 + 2.44948 + 2.82842 + 3.16228 + 3.4641 + 3.74166 + 2 = 18.64594$$

Step 11. Multiply by the front factor:

$$T_6 = 0.25 \times 18.64594 = 4.66148.$$

Step 12. Exact value: $\displaystyle\int_1^4 \sqrt{x}\,dx = \Big[\frac{2}{3}x^{3/2}\Big]_1^4 = \frac{2}{3}(8-1) = \frac{14}{3} = 4.666667.$

Step 13. Actual error: $\frac{14}{3} - T_6 = 4.666667 - 4.66148 = 0.005182$, which is positive ($T_6$ underestimates, since $f'' < 0$), and $0.005182 \le 0.015625$, so the bound is respected.

Answer. $E_T \le \dfrac{1}{64} = 0.015625$; $T_6 \approx 4.6615$, actual error 0.00518.

16. Find a bound on the error in approximating $\displaystyle\int_1^2 \frac{1}{x}\,dx$ using S_4, and compare it with the actual error found in Problem 8.

SOLUTION

Step 1. Identify: $f(x) = \dfrac{1}{x} = x^{-1}$, $a = 1$, $b = 2$, $n = 4$, and $E_S \le \dfrac{M(b-a)^5}{180n^4}$ with $|f^{(4)}(x)| \le M$ on $[1, 2]$.

Step 2. Differentiate four times: $f'(x) = -x^{-2}$, $f''(x) = 2x^{-3}$, $f^{(3)}(x) = -6x^{-4}$, $f^{(4)}(x) = 24x^{-5} = \dfrac{24}{x^5}$.

Step 3. Maximise $|f^{(4)}|$ on $[1, 2]$: $\dfrac{24}{x^5}$ is positive and decreasing for $x > 0$, so the maximum is at $x = 1$:

$$M = \frac{24}{1^5} = 24.$$

Step 4. Substitute $M = 24$, $b - a = 1$, $n = 4$:

$$E_S \le \frac{24 \cdot 1^5}{180 \cdot 4^4} = \frac{24}{180 \cdot 256} = \frac{24}{46080} = \frac{1}{1920} = 0.0005208.$$

Step 5. Compare with Problem 8: the actual error of S_4 there was $|\ln 2 - S_4| = 0.000107$, which is below 0.0005208. Compare also with the Trapezoidal bound for $n = 4$ on the same integral: with $K = 2$, $\dfrac{2 \cdot 1}{12 \cdot 16} = 0.010417$, about 20 times larger.

Answer. $E_S \le \dfrac{1}{1920} \approx 5.21 \times 10^{-4}$.

Choosing n for a required accuracy

17. How large must n be so that T_n approximates $\displaystyle\int_0^1 e^x\,dx$ with error less than 0.001?

SOLUTION

Step 1. Setup: $f(x) = e^x$ on $[0,1]$ with $f''(x) = e^x$, and we need $E_T \le \dfrac{K(b-a)^3}{12n^2} < 0.001$.

Step 2. The maximum of $|f''(x)| = e^x$ on $[0,1]$ is at $x = 1$ (increasing function), so $K = e \approx 2.718282$.

Step 3. Write the requirement:

$$\frac{2.718282 \cdot 1^3}{12n^2} < 0.001.$$

Step 4. Multiply by $12n^2 > 0$ and divide by 0.001:

$$n^2 > \frac{2.718282}{12 \times 0.001} = \frac{2.718282}{0.012} = 226.5235.$$

Step 5. Take the positive square root:

$$n > \sqrt{226.5235} \approx 15.0507.$$

(Check: $15^2 = 225 < 226.5235$ and $16^2 = 256 > 226.5235$.)

Step 6. The Trapezoidal Rule has no parity requirement, and n must be an integer strictly greater than 15.0507, so the smallest choice is $n = 16$.

Step 7. Verify: $n = 16$ gives $\dfrac{2.718282}{12 \cdot 256} = \dfrac{2.718282}{3072} = 8.848 \times 10^{-4} < 0.001$ (holds).

$n = 15$ gives $\dfrac{2.718282}{12 \cdot 225} = \dfrac{2.718282}{2700} = 1.0068 \times 10^{-3} > 0.001$ (fails).

Answer. $n = 16$.

18. How large must (even) n be so that S_n approximates $\displaystyle\int_0^2 x^4\,dx$ with error less than 0.01?

SOLUTION

Step 1. Setup: $f(x) = x^4$ on $[0,2]$, and we need $E_S \le \dfrac{M(b-a)^5}{180n^4} < 0.01$ with n even.

Step 2. Differentiate four times: $f'(x) = 4x^3$, $f''(x) = 12x^2$, $f^{(3)}(x) = 24x$, $f^{(4)}(x) = 24$. The fourth derivative is the constant 24, so $M = 24$.

Step 3. Write the requirement with $b - a = 2$:

$$\frac{24 \cdot 2^5}{180n^4} < 0.01 \implies \frac{768}{180n^4} < 0.01.$$

Step 4. Multiply by $n^4 > 0$ and divide by 0.01:

$$n^4 > \frac{768}{180 \times 0.01} = \frac{768}{1.8} = 426.667.$$

Step 5. Fourth root: $\sqrt{426.667} \approx 20.656$, then $\sqrt{20.656} \approx 4.545$, so $n > 4.545$.

Step 6. The smallest **even** integer greater than 4.545 is $n = 6$ (the next even integer after 4).

Step 7. Verify: $n = 6$ gives $\dfrac{768}{180 \cdot 1296} = \dfrac{768}{233280} = 0.003292 < 0.01$ (holds). $n = 4$ gives $\dfrac{768}{180 \cdot 256} = \dfrac{768}{46080} = 0.016667 > 0.01$ (fails).

Answer. $n = 6$.

19. How large must n be so that M_n approximates $\displaystyle\int_0^2 e^{-x^2}\,dx$ with error less than 10^{-4}? (Find K by maximising $|f''|$ on $[0, 2]$.)

SOLUTION

Step 1. Setup: $f(x) = e^{-x^2}$ on $[0, 2]$, and we need $E_M \le \dfrac{K(b-a)^3}{24n^2} < 10^{-4}$ with $|f''(x)| \le K$ on $[0, 2]$.

Step 2. First derivative (chain rule): $f'(x) = -2xe^{-x^2}$. Second derivative (product rule):

$$f''(x) = -2e^{-x^2} + (-2x)(-2xe^{-x^2}) = (4x^2 - 2)e^{-x^2}.$$

Step 3. To maximise $|f''|$ on $[0, 2]$, find the critical points of f'' using $f^{(3)}$:

$$f^{(3)}(x) = 8xe^{-x^2} + (4x^2 - 2)(-2x)e^{-x^2} = (12x - 8x^3)e^{-x^2} = 4x(3 - 2x^2)e^{-x^2}.$$

It vanishes at $x = 0$ and $x = \sqrt{1.5} \approx 1.2247$ (in $[0, 2]$).

Step 4. Evaluate f'' at the critical points and the endpoints:

$$f''(0) = -2, \quad f''(1.2247) = (6-2)e^{-1.5} = 4(0.223130) = 0.892521, \quad f''(2) = 14e^{-4} = 14(0.018316) = 0.256423.$$

The largest absolute value is $|f''(0)| = 2$, so $K = 2$.

Step 5. Write the requirement with $b - a = 2$:

$$\frac{2 \cdot 2^3}{24n^2} = \frac{16}{24n^2} = \frac{2}{3n^2} < 10^{-4}.$$

Step 6. Solve:

$$n^2 > \frac{2}{3 \times 10^{-4}} = 6666.67 \implies n > \sqrt{6666.67} \approx 81.65.$$

(Check: $81^2 = 6561 < 6666.67 < 6724 = 82^2$.)

Step 7. The smallest integer greater than 81.65 is $n = 82$. Verify: $n = 82$: $\dfrac{2}{3 \cdot 6724} = 9.915 \times 10^{-5} < 10^{-4}$ (holds); $n = 81$: $\dfrac{2}{3 \cdot 6561} = 1.0161 \times 10^{-4} > 10^{-4}$ (fails).

Answer. $K = 2$, $n = 82$.

20. How large must (even) n be so that Simpson's Rule S_n approximates $\displaystyle\int_1^3 \ln x\,dx$ with error less than 10^{-6}?

SOLUTION

Step 1. Setup: $f(x) = \ln x$ on $[1,3]$, and we need $E_S \le \dfrac{M(b-a)^5}{180n^4} < 10^{-6}$ with n even.

Step 2. Differentiate four times: $f'(x) = x^{-1}$, $f''(x) = -x^{-2}$, $f^{(3)}(x) = 2x^{-3}$, $f^{(4)}(x) = -6x^{-4} = -\dfrac{6}{x^4}$.

Step 3. Maximise $|f^{(4)}|$ on $[1,3]$: $|f^{(4)}(x)| = \dfrac{6}{x^4}$ is decreasing for $x > 0$, so the maximum is at $x = 1$: $M = 6$.

Step 4. Write the requirement with $b - a = 2$:

$$\frac{6 \cdot 2^5}{180n^4} = \frac{192}{180n^4} = \frac{16}{15n^4} < 10^{-6}.$$

Step 5. Solve:

$$n^4 > \frac{16}{15 \times 10^{-6}} = 1\,066\,667 \implies n > (1\,066\,667)^{1/4}.$$

Since $\sqrt{1\,066\,667} \approx 1032.8$ and $\sqrt{1032.8} \approx 32.14$, we get $n > 32.14$.

Step 6. The smallest **even** integer greater than 32.14 is $n = 34$. Verify: $34^4 = 1\,336\,336$, so $\dfrac{16}{15 \cdot 1\,336\,336} = 7.98 \times 10^{-7} < 10^{-6}$ (holds). For $n = 32$: $32^4 = 1\,048\,576$ and $\dfrac{16}{15 \cdot 1\,048\,576} = 1.0173 \times 10^{-6} > 10^{-6}$ (fails, just barely).

Answer. $M = 6$, $n = 34$.

21. For $\displaystyle\int_0^1 e^x\,dx$, find the smallest n that guarantees an error below 10^{-6} for (a) the Trapezoidal Rule T_n and (b) Simpson's Rule S_n (even n). Compare the number of function evaluations ($n + 1$ in each case).

SOLUTION

Step 1. For $f(x) = e^x$ on $[0,1]$, every derivative is e^x, which is increasing, so $K = M = e \approx 2.718282$ for both the second and the fourth derivative.

Step 2. Trapezoidal Rule. Require $\dfrac{e \cdot 1^3}{12n^2} < 10^{-6}$:

$$n^2 > \frac{2.718282}{12 \times 10^{-6}} = 226\,523.5 \implies n > \sqrt{226\,523.5} \approx 475.9.$$

(Check: $475^2 = 225\,625 < 226\,523.5 < 226\,576 = 476^2$.) So $n = 476$.

Step 3. Simpson's Rule. Require $\dfrac{e \cdot 1^5}{180n^4} < 10^{-6}$:

$$n^4 > \frac{2.718282}{180 \times 10^{-6}} = 15\,101.6 \implies n > (15\,101.6)^{1/4} \approx 11.09.$$

(Check: $11^4 = 14\,641 < 15\,101.6 < 20\,736 = 12^4$.) The smallest even integer above 11.09 is $n = 12$.

Step 4. Count function evaluations: a rule with n subintervals uses $n + 1$ grid values for T_n and S_n.

$$T_{476}: \ 477 \text{ values}, \qquad S_{12}: \ 13 \text{ values}.$$

Step 5. Ratio: $\dfrac{477}{13} \approx 36.7$, so Simpson's Rule needs about 37 times fewer function evaluations for the same guaranteed accuracy, because its error bound decreases like n^{-4} instead of n^{-2}.

Answer. T_{476} (477 values) versus S_{12} (13 values); about 37 times fewer evaluations for Simpson.

Concavity, comparison of the rules and exact values

22. Without computing any sum, use the sign of f'' to decide whether T_n and M_n overestimate or underestimate each integral. Then confirm with T_4 and M_4. (a) $\displaystyle\int_1^3 \frac{1}{x}\,dx$. (b) $\displaystyle\int_1^5 \ln x\,dx$. (c) $\displaystyle\int_0^2 x^3\,dx$. (d) $\displaystyle\int_0^\pi \sin x\,dx$.

SOLUTION

Step 1. (a) $f(x) = \dfrac{1}{x}$ **on** $[1, 3]$**.** $f'(x) = -x^{-2}$ and $f''(x) = \dfrac{2}{x^3} > 0$ on $[1, 3]$, so f is concave up: T_n **overestimates** and M_n **underestimates**.

Step 2. Numerical check with $n = 4$ (exact value $\ln 3 = 1.09861$). Trapezoidal: with $\Delta x = 0.5$, the nodes are at $x = 1,\ 1.5,\ 2,\ 2.5,\ 3$, and the function values are

$$f(1) = 1 \qquad f(1.5) = 0.666667$$
$$f(2) = 0.5 \qquad f(2.5) = 0.4$$
$$f(3) = 0.333333$$

$$T_4 = 0.25\Big[\,1 + 2(0.666667) + 2(0.5) + 2(0.4) + 0.333333\,\Big]$$
$$T_4 = 0.25 \times 4.466667 = 1.11667.$$

Midpoint: with $\Delta x = 0.5$, the midpoints are at $x = 1.25,\ 1.75,\ 2.25,\ 2.75$, and the function values are

$$f(1.25) = 0.8 \qquad f(1.75) = 0.571429$$
$$f(2.25) = 0.444444 \qquad f(2.75) = 0.363636$$

$$M_4 = 0.5\Big[\,0.8 + 0.571429 + 0.444444 + 0.363636\,\Big]$$
$$M_4 = 0.5 \times 2.179509 = 1.08975.$$

Step 3. Result: $T_4 - I = 0.0181 > 0$ and $M_4 - I = -0.00886 < 0$, confirming the prediction.

Step 4. (b) $f(x) = \ln x$ **on** $[1, 5]$**.** $f'(x) = \dfrac{1}{x}$ and $f''(x) = -\dfrac{1}{x^2} < 0$, so f is concave down: T_n **underestimates** and M_n **overestimates**.

Step 5. Numerical check with $n = 4$ (exact value $\int_1^5 \ln x\,dx = [x\ln x - x]_1^5 = 5\ln 5 - 4 = 4.04719$). Trapezoidal: with $\Delta x = 1$, the nodes are at $x = 1,\ 2,\ 3,\ 4,\ 5$, and the function values are

$$f(1) = 0 \qquad f(2) = 0.693147$$
$$f(3) = 1.09861 \qquad f(4) = 1.38629$$
$$f(5) = 1.60944$$
$$T_4 = 0.5\Big[\,0 + 2(0.693147) + 2(1.09861) + 2(1.38629) + 1.60944\,\Big]$$
$$T_4 = 0.5 \times 7.965534 = 3.98277.$$

Midpoint: with $\Delta x = 1$, the midpoints are at $x = 1.5,\ 2.5,\ 3.5,\ 4.5$, and the function values are

$$f(1.5) = 0.405465 \qquad f(2.5) = 0.916291$$
$$f(3.5) = 1.25276 \qquad f(4.5) = 1.50408$$
$$M_4 = 1\Big[\,0.405465 + 0.916291 + 1.25276 + 1.50408\,\Big]$$
$$M_4 = 1 \times 4.078596 = 4.0786.$$

Step 6. Result: $T_4 - I = -0.0644 < 0$ and $M_4 - I = +0.0314 > 0$, confirming the prediction.

Step 7. (c) $f(x) = x^3$ **on** $[0, 2]$**.** $f'(x) = 3x^2$ and $f''(x) = 6x \geq 0$ on $[0, 2]$ (strictly positive for $x > 0$), so f is concave up: T_n **overestimates** and M_n **underestimates**.

Step 8. Numerical check with $n = 4$ (exact value $\int_0^2 x^3\,dx = \left[\dfrac{x^4}{4}\right]_0^2 = 4$). Trapezoidal: with $\Delta x = 0.5$, the nodes are at $x = 0,\ 0.5,\ 1,\ 1.5,\ 2$, and the function values are

$$f(0) = 0 \qquad f(0.5) = 0.125$$
$$f(1) = 1 \qquad f(1.5) = 3.375$$
$$f(2) = 8$$
$$T_4 = 0.25\Big[\,0 + 2(0.125) + 2(1) + 2(3.375) + 8\,\Big]$$
$$T_4 = 0.25 \times 17 = 4.25.$$

Midpoint: with $\Delta x = 0.5$, the midpoints are at $x = 0.25,\ 0.75,\ 1.25,\ 1.75$, and the function values are

$$f(0.25) = 0.015625 \qquad f(0.75) = 0.421875$$
$$f(1.25) = 1.95312 \qquad f(1.75) = 5.35938$$
$$M_4 = 0.5\Big[\,0.015625 + 0.421875 + 1.95312 + 5.35938\,\Big]$$
$$M_4 = 0.5 \times 7.75 = 3.875.$$

Step 9. Result: $T_4 - 4 = 0.25 > 0$ and $M_4 - 4 = -0.125 < 0$, confirming the prediction.

Step 10. (d) $f(x) = \sin x$ **on** $[0, \pi]$**.** $f'(x) = \cos x$ and $f''(x) = -\sin x \leq 0$ on $[0, \pi]$ (since $\sin x \geq 0$ there), so f is concave down: T_n **underestimates** and M_n **overestimates**.

Step 11. Numerical check with $n = 4$ (exact value $\int_0^\pi \sin x\,dx = 2$; here $\Delta x = \dfrac{\pi}{4} =$

0.785398). Trapezoidal: with $\Delta x = 0.785398$, the nodes are at $x = 0,\ \frac{\pi}{4},\ \frac{\pi}{2},\ \frac{3\pi}{4},\ \pi$, and the function values are

$$f(0) = 0 \qquad f(\frac{\pi}{4}) = 0.707107$$
$$f(\frac{\pi}{2}) = 1 \qquad f(\frac{3\pi}{4}) = 0.707107$$
$$f(\pi) = 0$$
$$T_4 = 0.392699\Big[\,0 + 2(0.707107) + 2(1) + 2(0.707107) + 0\,\Big]$$
$$T_4 = 0.392699 \times 4.828428 = 1.89612.$$

Midpoint: with $\Delta x = 0.785398$, the midpoints are at $x = 0.392699,\ 1.1781,\ 1.9635,\ 2.74889$, and the function values are

$$f(0.392699) = 0.382683 \qquad f(1.1781) = 0.92388$$
$$f(1.9635) = 0.92388 \qquad f(2.74889) = 0.382683$$
$$M_4 = 0.785398\Big[\,0.382683 + 0.92388 + 0.92388 + 0.382683\,\Big]$$
$$M_4 = 0.785398 \times 2.613126 = 2.05234.$$

Step 12. Result: $T_4 - 2 = -0.104 < 0$ and $M_4 - 2 = +0.0523 > 0$, confirming the prediction.

Answer. (a) T over, M under; (b) T under, M over; (c) T over, M under; (d) T under, M over (concave up gives $M_n \le I \le T_n$, concave down gives $T_n \le I \le M_n$).

23. A student computes T_4 and M_4 for the same integral and gets $T_4 = 5.20$ and $M_4 = 5.05$. Without further calculation, give the best available estimate for the Simpson value with 8 subintervals using the identity $S_{2n} = \dfrac{2M_n + T_n}{3}$, and explain why this weighted average reduces the error compared with either T_4 or M_4 alone.

SOLUTION

Step 1. The identity connecting the three rules is $S_{2n} = \dfrac{2M_n + T_n}{3}$. Here $M_n = M_4$ and $T_n = T_4$ (so $n = 4$), hence the Simpson value obtained is for $2n = 8$ subintervals: S_8.

Step 2. Substitute $T_4 = 5.20$ and $M_4 = 5.05$:

$$S_8 = \frac{2(5.05) + 5.20}{3}.$$

Step 3. Evaluate the numerator: $2(5.05) = 10.10$ and $10.10 + 5.20 = 15.30$. Divide:

$$S_8 = \frac{15.30}{3} = 5.10.$$

Step 4. Independent check by another route. The Trapezoidal Rule with doubled n is the average $T_{2n} = \dfrac{T_n + M_n}{2}$, so $T_8 = \dfrac{5.20 + 5.05}{2} = 5.125$. Simpson's Rule can also be written $S_{2n} = \dfrac{4T_{2n} - T_n}{3}$, which gives

$$S_8 = \frac{4(5.125) - 5.20}{3} = \frac{20.5 - 5.2}{3} = \frac{15.3}{3} = 5.10,$$

the same value.

Step 5. Meaning of the data: $T_4 = 5.20 > M_4 = 5.05$ is the pattern of a concave-up integrand (M under, T over), so the exact integral lies between 5.05 and 5.20, and 5.10 is in that interval.

Step 6. Why the $2:1$ weighting reduces the error: for a smooth integrand the leading error terms satisfy $I - M_n \approx e$ and $I - T_n \approx -2e$ (same size ratio as the bound constants $\frac{1}{24}$ and $\frac{1}{12}$, opposite signs). Then

$$I - \frac{2M_n + T_n}{3} \approx \frac{2e + (-2e)}{3} = 0,$$

so the leading term cancels. With the given numbers, $T_4 - M_4 = 3e = 0.15$ gives $e = 0.05$ and $I \approx M_4 + e = 5.10$, matching S_8.

Step 7. An equal-weight average does not cancel the leading term: $I - \frac{T_n + M_n}{2} \approx \frac{-2e + e}{2} = -\frac{e}{2} \neq 0$, so $\frac{T_4 + M_4}{2} = 5.125$ is a worse estimate than 5.10 (and it is T_8, not S_8).

Answer. $S_8 = \frac{2(5.05) + 5.20}{3} = 5.10.$

24. For $\int_1^2 \frac{1}{x}\,dx$, compute T_2 and M_2, then use $S_{2n} = \frac{2M_n + T_n}{3}$ to obtain S_4. Compare with the value of S_4 computed directly in Problem 8.

SOLUTION

Step 1. Compute T_2 for $\int_1^2 \frac{dx}{x}$.

Step 2. Formula used (Trapezoidal Rule, $n = 2$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 2$ this is

$$T_2 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + f(x_2)\Big].$$

Step 3. Width of each subinterval:

$$\Delta x = \frac{b - a}{n} = \frac{2 - 1}{2} = 0.5.$$

Step 4. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 2$:

$$x_0 = 1 \qquad x_1 = 1.5 \qquad x_2 = 2$$

Step 5. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = \frac{1}{1} = 1 \qquad f(x_1) = \frac{1}{1.5} = 0.666667$$
$$f(x_2) = \frac{1}{2} = 0.5$$

Step 6. Insert the weights $(1, 2, 2, \dots, 2, 1)$ and the function values:

$$T_2 = \frac{0.5}{2}\Big[\,1 + 2(0.666667) + 0.5\,\Big]$$

Step 7. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; 1 + 1.333334 + 0.5 \; = \; 2.833334$$

Step 8. Multiply by the front factor:

$$T_2 = 0.25 \times 2.833334 = 0.708333.$$

Step 9. Compute M_2.

Step 10. Formula used (Midpoint Rule, $n = 2$):

$$M_n = \Delta x \left[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)\right].$$

For $n = 2$ this is

$$M_2 = \Delta x\Big[f(\bar{x}_1) + f(\bar{x}_2)\Big].$$

Step 11. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{2-1}{2} = 0.5.$$

Step 12. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$:

$$\bar{x}_1 = \frac{1 + 1.5}{2} = 1.25 \qquad \bar{x}_2 = \frac{1.5 + 2}{2} = 1.75$$

Step 13. Function values at the midpoints (rounded to 6 significant figures where necessary):

$$f(\bar{x}_1) = \frac{1}{1.25} = 0.8 \qquad f(\bar{x}_2) = \frac{1}{1.75} = 0.571429$$

Step 14. Add the 2 midpoint values (all weights are 1):

$$\text{bracket} = \; 0.8 + 0.571429 \; = \; 1.371429$$

Step 15. Multiply by Δx:

$$M_2 = 0.5 \times 1.371429 = 0.685715.$$

Step 16. Combine with $S_{2n} = \dfrac{2M_n + T_n}{3}$ for $n = 2$:

$$S_4 = \frac{2(0.685715) + 0.708333}{3} = \frac{1.37143 + 0.708333}{3} = \frac{2.079763}{3} = 0.693254.$$

Step 17. Compare with the direct Simpson computation in Problem 8: $S_4 = 0.693254$. The two agree (to the rounding used), confirming the identity for this integral.

Answer. $T_2 \approx 0.708333$, $M_2 \approx 0.685715$, so $S_4 = \dfrac{2M_2 + T_2}{3} \approx 0.693254$, equal to the direct value.

25. Explain, using the error-bound formulas, why doubling n reduces the Trapezoidal error bound by a factor of 4 but reduces the Simpson error bound by a factor of 16.

SOLUTION

Step 1. Write the bounds with their n-dependence:

$$B_T(n) = \frac{K(b-a)^3}{12n^2}, \qquad B_S(n) = \frac{M(b-a)^5}{180n^4}.$$

($B_M(n) = \dfrac{K(b-a)^3}{24n^2}$ behaves like B_T.)

Step 2. Replace n by $2n$ in the Trapezoidal bound:

$$B_T(2n) = \frac{K(b-a)^3}{12(2n)^2} = \frac{K(b-a)^3}{12 \cdot 4n^2} = \frac{1}{4} \cdot \frac{K(b-a)^3}{12n^2} = \frac{1}{4} B_T(n).$$

Since $(2n)^2 = 4n^2$, the bound is divided by 4.

Step 3. Replace n by $2n$ in the Simpson bound:

$$B_S(2n) = \frac{M(b-a)^5}{180(2n)^4} = \frac{M(b-a)^5}{180 \cdot 16n^4} = \frac{1}{16} \cdot \frac{M(b-a)^5}{180n^4} = \frac{1}{16} B_S(n).$$

Since $(2n)^4 = 16n^4$, the bound is divided by 16.

Step 4. Numerical illustration for $\displaystyle\int_1^2 \frac{dx}{x}$ (with $K = 2$ and $M = 24$ from Example 4 and Problem 16):

$$B_T(10) = \frac{2}{12 \cdot 100} = \frac{1}{600} = 0.001667, \quad B_T(20) = \frac{2}{12 \cdot 400} = \frac{1}{2400} = 0.0004167, \quad \frac{B_T(10)}{B_T(20)} = 4,$$

$$B_S(4) = \frac{24}{180 \cdot 256} = \frac{1}{1920} = 0.0005208, \quad B_S(8) = \frac{24}{180 \cdot 4096} = \frac{1}{30720} = 0.00003255, \quad \frac{B_S(4)}{B_S(8)} = 16.$$

Step 5. Consequence: to gain one more decimal digit of accuracy (error divided by 10), the Trapezoidal Rule needs n multiplied by $\sqrt{10} \approx 3.16$, whereas Simpson's Rule needs n multiplied only by $10^{1/4} \approx 1.78$. This is why Simpson's Rule reaches high accuracy with far fewer function evaluations.

Answer. $B_T(2n) = \dfrac{1}{4} B_T(n)$ because $(2n)^2 = 4n^2$; $B_S(2n) = \dfrac{1}{16} B_S(n)$ because $(2n)^4 = 16n^4$.

26. For $\displaystyle\int_1^2 \frac{dx}{x}$, compute T_2, T_4, S_2, S_4, the actual error of each against $\ln 2$, and the ratios $\dfrac{|I-T_2|}{|I-T_4|}$ and $\dfrac{|I-S_2|}{|I-S_4|}$. Compare them with the factors 4 and 16 predicted by the error bounds.

SOLUTION

Step 1. Exact value: $\displaystyle\int_1^2 \frac{dx}{x} = \ln 2 = 0.693147.$

Step 2. Trapezoidal Rule, $n = 2$: with $\Delta x = 0.5$, the nodes are at $x = 1,\ 1.5,\ 2$, and the function values are

$$f(1) = 1 \qquad f(1.5) = 0.666667$$
$$f(2) = 0.5$$
$$T_2 = 0.25\Big[\,1 + 2(0.666667) + 0.5\,\Big]$$
$$T_2 = 0.25 \times 2.833334 = 0.708333.$$

Step 3. Trapezoidal Rule, $n = 4$: with $\Delta x = 0.25$, the nodes are at $x = 1,\ 1.25,\ 1.5,\ 1.75,\ 2$, and the function values are

$$f(1) = 1 \qquad f(1.25) = 0.8$$
$$f(1.5) = 0.666667 \qquad f(1.75) = 0.571429$$
$$f(2) = 0.5$$
$$T_4 = 0.125\Big[\,1 + 2(0.8) + 2(0.666667) + 2(0.571429) + 0.5\,\Big]$$
$$T_4 = 0.125 \times 5.576192 = 0.697024.$$

Step 4. Errors and ratio:

$$|I-T_2| = 0.01519, \qquad |I-T_4| = 0.003877, \qquad \frac{|I-T_2|}{|I-T_4|} = \frac{0.01519}{0.003877} = 3.92.$$

Doubling n cut the error by a factor of about 4, as the n^{-2} behaviour predicts.

Step 5. Simpson's Rule, $n = 2$: with $\Delta x = 0.5$, the nodes are at $x = 1,\ 1.5,\ 2$, and the function values are

$$f(1) = 1 \qquad f(1.5) = 0.666667$$
$$f(2) = 0.5$$
$$S_2 = 0.166667\Big[\,1 + 4(0.666667) + 0.5\,\Big]$$
$$S_2 = 0.166667 \times 4.166668 = 0.694445.$$

Step 6. Simpson's Rule, $n = 4$: with $\Delta x = 0.25$, the nodes are at $x = 1,\ 1.25,\ 1.5,\ 1.75,\ 2$, and the function values are

$$f(1) = 1 \qquad f(1.25) = 0.8$$
$$f(1.5) = 0.666667 \qquad f(1.75) = 0.571429$$
$$f(2) = 0.5$$
$$S_4 = 0.0833333\Big[\,1 + 4(0.8) + 2(0.666667) + 4(0.571429) + 0.5\,\Big]$$
$$S_4 = 0.0833333 \times 8.31905 = 0.693254.$$

Step 7. Errors and ratio:

$$|I - S_2| = 0.001297, \qquad |I - S_4| = 0.0001068, \qquad \frac{|I - S_2|}{|I - S_4|} = \frac{0.001297}{0.0001068} = 12.1.$$

The error fell by a factor of about 12. The bound predicts a factor of 16 in the worst case; at such small n the observed factor is still below 16, and it approaches 16 as n grows (a further computer check gives $|I - S_8| = 0.00000735$, so $|I - S_4|/|I - S_8| \approx 14.5$).

Step 8. Comparison: with the same data ($n = 4$, five function values) Simpson's error 0.000107 is about 36 times smaller than the Trapezoidal error 0.00388.

Answer. Error ratios ≈ 3.92 (Trapezoidal) and ≈ 12.1 (Simpson), compared with the predicted factors 4 and 16.

27. The integral $\displaystyle\int_0^1 \frac{4}{1+x^2}\,dx$ equals π. Approximate it using T_4, M_4 and S_4, compute the actual error of each, and rank the three rules from most to least accurate.

SOLUTION

Step 1. Exact value: $\displaystyle\int_0^1 \frac{4}{1+x^2}\,dx = 4\Big[\arctan x\Big]_0^1 = 4\left(\frac{\pi}{4} - 0\right) = \pi \approx 3.141593.$

Step 2. Trapezoidal Rule T_4.

Step 3. Formula used (Trapezoidal Rule, $n = 4$):

$$T_n = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\Big].$$

For $n = 4$ this is

$$T_4 = \frac{\Delta x}{2}\Big[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)\Big].$$

Step 4. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{4} = 0.25.$$

Step 5. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 0 \qquad x_1 = 0.25 \qquad x_2 = 0.5 \qquad x_3 = 0.75 \qquad x_4 = 1$$

Step 6. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0) = \frac{4}{1+(0)^2} = 4 \qquad f(x_1) = \frac{4}{1+(0.25)^2} = 3.76471$$

$$f(x_2) = \frac{4}{1+(0.5)^2} = 3.2 \qquad f(x_3) = \frac{4}{1+(0.75)^2} = 2.56$$

$$f(x_4) = \frac{4}{1+(1)^2} = 2$$

Step 7. Insert the weights $(1, 2, 2, \ldots, 2, 1)$ and the function values:

$$T_4 = \frac{0.25}{2}\Big[\,4 + 2(3.76471) + 2(3.2) + 2(2.56) + 2\,\Big]$$

Step 8. Do each multiplication inside the bracket and add the results:

$$\text{bracket} = \; 4 + 7.52942 + 6.4 + 5.12 + 2 \; = \; 25.04942$$

Step 9. Multiply by the front factor:

$$T_4 = 0.125 \times 25.04942 = 3.13118.$$

Step 10. Midpoint Rule M_4.

Step 11. Formula used (Midpoint Rule, $n = 4$):

$$M_n = \Delta x \left[f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n) \right].$$

For $n = 4$ this is

$$M_4 = \Delta x \Big[f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) + f(\bar{x}_4) \Big].$$

Step 12. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{4} = 0.25.$$

Step 13. Midpoints of the subintervals, $\bar{x}_i = \dfrac{x_{i-1} + x_i}{2}$:

$$\bar{x}_1 = \frac{0 + 0.25}{2} = 0.125 \qquad \bar{x}_2 = \frac{0.25 + 0.5}{2} = 0.375$$

$$\bar{x}_3 = \frac{0.5 + 0.75}{2} = 0.625 \qquad \bar{x}_4 = \frac{0.75 + 1}{2} = 0.875$$

Step 14. Function values at the midpoints (rounded to 6 significant figures where necessary):

$$f(\bar{x}_1) = \frac{4}{1 + (0.125)^2} = 3.93846 \qquad f(\bar{x}_2) = \frac{4}{1 + (0.375)^2} = 3.50685$$

$$f(\bar{x}_3) = \frac{4}{1 + (0.625)^2} = 2.8764 \qquad f(\bar{x}_4) = \frac{4}{1 + (0.875)^2} = 2.26549$$

Step 15. Add the 4 midpoint values (all weights are 1):

$$\text{bracket} = \ 3.93846 + 3.50685 + 2.8764 + 2.26549 \ = \ 12.5872$$

Step 16. Multiply by Δx:

$$M_4 = 0.25 \times 12.5872 = 3.1468.$$

Step 17. Simpson's Rule S_4.

Step 18. Formula used (Simpson's Rule, $n = 4$):

$$S_n = \frac{\Delta x}{3} \Big[f(x_0) + 4f(x_1) + 2f(x_2) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n) \Big].$$

For $n = 4$ this is

$$S_4 = \frac{\Delta x}{3} \Big[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4) \Big].$$

Step 19. Width of each subinterval:

$$\Delta x = \frac{b-a}{n} = \frac{1-0}{4} = 0.25.$$

Step 20. Grid points $x_i = a + i\,\Delta x$ for $i = 0, 1, \ldots, 4$:

$$x_0 = 0 \qquad x_1 = 0.25 \qquad x_2 = 0.5 \qquad x_3 = 0.75 \qquad x_4 = 1$$

Step 21. Function values at the grid points (rounded to 6 significant figures where necessary):

$$f(x_0)=\frac{4}{1+(0)^2}=4 \qquad f(x_1)=\frac{4}{1+(0.25)^2}=3.76471$$
$$f(x_2)=\frac{4}{1+(0.5)^2}=3.2 \qquad f(x_3)=\frac{4}{1+(0.75)^2}=2.56$$
$$f(x_4)=\frac{4}{1+(1)^2}=2$$

Step 22. Insert the weights $(1,4,2,4,\dots,2,4,1)$ and the function values:
$$S_4=\frac{0.25}{3}\Big[\,4+4(3.76471)+2(3.2)+4(2.56)+2\,\Big]$$

Step 23. Do each multiplication inside the bracket and add the results:
$$\text{bracket}=\ 4+15.05884+6.4+10.24+2\ =\ 37.69884$$

Step 24. Multiply by the front factor:
$$S_4=\frac{0.25}{3}\,(\approx 0.0833333)\times 37.69884=3.14157.$$

Step 25. Errors (computed from unrounded function values, so they may differ slightly from the differences of the rounded results above):
$$T_4-\pi=-0.01042, \qquad M_4-\pi=+0.005208, \qquad S_4-\pi=-0.000024.$$

Step 26. Ranking by absolute error: S_4 $(0.000024)<M_4$ $(0.00521)<T_4$ (0.0104). Simpson's value 3.14157 agrees with π to 3 decimal places (3.1416 when rounded to 4 places) using only five function values. Also $|\pi-T_4|\approx 2|M_4-\pi|$ $(0.0104\approx 2\times 0.00521)$.

Answer. $T_4\approx 3.13118$, $M_4\approx 3.1468$, $S_4\approx 3.14157$ against $\pi\approx 3.141593$; ranking S_4, M_4, T_4.

28. Let $f(x)=\ln x$ on $[1,3]$ with $n=4$. Without computing, put L_4, R_4, T_4, M_4 and the exact integral I in increasing order, justifying each inequality from the monotonicity and concavity of f; then confirm numerically. (L_4 and R_4 are the left- and right-endpoint Riemann sums.)

SOLUTION

Step 1. Recall the endpoint sums for $\Delta x=\dfrac{b-a}{n}$: $L_n=\Delta x\big[f(x_0)+f(x_1)+\cdots+f(x_{n-1})\big]$ (left endpoints) and $R_n=\Delta x\big[f(x_1)+\cdots+f(x_n)\big]$ (right endpoints). Adding them, the interior points appear twice and the end points once, so $T_n=\dfrac{L_n+R_n}{2}$.

Step 2. Order from the properties of $f(x)=\ln x$ on $[1,3]$: $f'(x)=\dfrac{1}{x}>0$, so f is **increasing**; therefore on each subinterval the left-endpoint height is the smallest and the right-endpoint height the largest, giving $L_n<I<R_n$ and $L_n<T_n<R_n$ (the average of two numbers lies between them). Also each midpoint height is less than the right-endpoint height, so $M_n<R_n$. And $f''(x)=-\dfrac{1}{x^2}<0$, so f is **concave down**, giving $T_n<I<M_n$.

Step 3. Combine: $L_n < T_n < I < M_n < R_n$. Now confirm numerically with $n = 4$ ($\Delta x = 0.5$, nodes $1, 1.5, 2, 2.5, 3$): $\ln 1 = 0$, $\ln 1.5 = 0.405465$, $\ln 2 = 0.693147$, $\ln 2.5 = 0.916291$, $\ln 3 = 1.098612$.

Step 4. Left sum: $L_4 = 0.5\,[0{+}0.405465{+}0.693147{+}0.916291] = 0.5{\times}2.014903 = 1.00745$. Right sum: $R_4 = 0.5\,[0.405465 + 0.693147 + 0.916291 + 1.098612] = 0.5 \times 3.113515 = 1.55676$.

Step 5. Trapezoidal Rule: with $\Delta x = 0.5$, the nodes are at $x = 1,\ 1.5,\ 2,\ 2.5,\ 3$, and the function values are

$$f(1) = 0 \qquad f(1.5) = 0.405465$$
$$f(2) = 0.693147 \qquad f(2.5) = 0.916291$$
$$f(3) = 1.09861$$

$$T_4 = 0.25\Big[\,0 + 2(0.405465) + 2(0.693147) + 2(0.916291) + 1.09861\,\Big]$$
$$T_4 = 0.25 \times 5.128416 = 1.2821.$$

Step 6. Midpoint Rule: with $\Delta x = 0.5$, the midpoints are at $x = 1.25,\ 1.75,\ 2.25,\ 2.75$, and the function values are

$$f(1.25) = 0.223144 \qquad f(1.75) = 0.559616$$
$$f(2.25) = 0.81093 \qquad f(2.75) = 1.0116$$

$$M_4 = 0.5\Big[\,0.223144 + 0.559616 + 0.81093 + 1.0116\,\Big]$$
$$M_4 = 0.5 \times 2.60529 = 1.30265.$$

Step 7. Exact value: $\displaystyle\int_1^3 \ln x\,dx = \Big[x\ln x{-}x\Big]_1^3 = (3\ln 3{-}3){-}(0{-}1) = 3\ln 3{-}2 = 1.29584$.

Step 8. Ordering:
$L_4 = 1.00745 < T_4 = 1.2821 < I = 1.29584 < M_4 = 1.30265 < R_4 = 1.55676$.
(Also $T_4 = \dfrac{L_4 + R_4}{2} = \dfrac{1.00745 + 1.55676}{2} = 1.2821$.)

Answer. $L_4 < T_4 < I < M_4 < R_4$, i.e. $1.0075 < 1.2821 < 1.2958 < 1.3026 < 1.5568$.

Error diagnosis

29. A student approximates $\displaystyle\int_0^2 x^2\,dx$ with S_4 and writes

$$S_4 = \frac{0.5}{3}\big[f(0) + 2f(0.5) + 4f(1) + 2f(1.5) + f(2)\big] = \frac{13}{6} \approx 2.167.$$

The student says that since Simpson's Rule is exact for polynomials of degree at most 3, the rule must be wrong for this problem. Find the error and give the correct result.

SOLUTION

Step 1. Recall the Simpson weights for $n = 4$: the end points get 1 and the interior points alternate $4, 2, 4$, so the bracket is $f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)$. The student used $2, 4, 2$ instead.

Step 2. Check the student's arithmetic: $f(0) = 0$, $f(0.5) = 0.25$, $f(1) = 1$, $f(1.5) = 2.25$, $f(2) = 4$. Student's bracket: $0 + 2(0.25) + 4(1) + 2(2.25) + 4 = 0 + 0.5 + 4 + 4.5 + 4 = 13$, and $\dfrac{0.5}{3} \times 13 = \dfrac{13}{6} = 2.1667$.

Step 3. Correct bracket: $f(0) + 4f(0.5) + 2f(1) + 4f(1.5) + f(2) = 0 + 4(0.25) + 2(1) + 4(2.25) + 4 = 0 + 1 + 2 + 9 + 4 = 16$.

Step 4. Correct value: $S_4 = \dfrac{0.5}{3} \times 16 = \dfrac{8}{3} = 2.6667$.

Step 5. Check against the exact value: $\displaystyle\int_0^2 x^2\,dx = \frac{8}{3}$. The correct S_4 equals it, as expected because Simpson's Rule is exact for x^2 (degree ≤ 3). The student's conclusion that the rule "is broken" is wrong: the rule was applied with the wrong weights, and 2.1667 is far from $\dfrac{8}{3}$ (-0.5 error).

Answer. Wrong weights ($2, 4, 2$ instead of $4, 2, 4$); correct $S_4 = \dfrac{8}{3}$, exact.

30. To make the Midpoint Rule M_n approximate $\displaystyle\int_0^1 e^x\,dx$ with error less than 0.001, a student writes

$$\frac{e\cdot 1^3}{12n^2} < 0.001 \implies n^2 > 226.5 \implies n = 16.$$

Find the error and give the correct smallest n.

SOLUTION

Step 1. The student used the denominator 12 in $\dfrac{K(b-a)^3}{12n^2}$, which is the **Trapezoidal** bound. The Midpoint bound is $E_M \le \dfrac{K(b-a)^3}{24n^2}$.

Step 2. Redo the calculation with $f(x) = e^x$, $K = e \approx 2.718282$ (since $f''(x) = e^x$ is increasing, its maximum on $[0,1]$ is at $x = 1$):

$$\frac{2.718282\cdot 1^3}{24n^2} < 0.001.$$

Step 3. Solve:

$$n^2 > \frac{2.718282}{24\times 0.001} = \frac{2.718282}{0.024} = 113.262 \implies n > \sqrt{113.262} \approx 10.64.$$

Step 4. The smallest integer greater than 10.64 is $n = 11$. Verify: $n = 11$: $\dfrac{2.718282}{24\cdot 121} = \dfrac{2.718282}{2904} = 9.360\times 10^{-4} < 0.001$ (holds); $n = 10$: $\dfrac{2.718282}{24\cdot 100} = 1.1326\times 10^{-3} > 0.001$ (fails).

Step 5. Assessment of the student's answer: $n = 16$ does guarantee the accuracy (its bound with the correct constant is $\dfrac{2.718282}{24\cdot 256} = 4.42\times 10^{-4} < 0.001$), but it is not the smallest n; the question asks how large n must be, and the correct minimum is $n = 11$ (the student's calculation actually finds the minimum for the Trapezoidal Rule).

Answer. The student used the Trapezoidal constant 12; with 24 the smallest n is 11.

31. A student claims: "For a positive function, the Trapezoidal Rule always overestimates the integral, because the trapezoids stick out above the curve." Find the error by computing T_4 for $\displaystyle\int_0^{\pi} \sin x\,dx$, and state the correct rule.

SOLUTION

Step 1. The claim is false; the direction of the Trapezoidal error depends on concavity, not on the sign of f. A single counterexample suffices. Take $f(x) = \sin x \geq 0$ on $[0, \pi]$.

Step 2. Compute T_4 ($\Delta x = \dfrac{\pi}{4}$): with $\Delta x = 0.785398$, the nodes are at $x = 0,\ \dfrac{\pi}{4},\ \dfrac{\pi}{2},\ \dfrac{3\pi}{4},\ \pi$, and the function values are

$$f(0) = 0 \qquad f(\tfrac{\pi}{4}) = 0.707107$$
$$f(\tfrac{\pi}{2}) = 1 \qquad f(\tfrac{3\pi}{4}) = 0.707107$$
$$f(\pi) = 0$$
$$T_4 = 0.392699\Big[\,0 + 2(0.707107) + 2(1) + 2(0.707107) + 0\,\Big]$$
$$T_4 = 0.392699 \times 4.828428 = 1.89612.$$

Step 3. The exact value is $\displaystyle\int_0^{\pi} \sin x\,dx = \Big[-\cos x\Big]_0^{\pi} = 2$. Since $T_4 = 1.89612 < 2$, the Trapezoidal Rule **underestimates** here even though $f \geq 0$.

Step 4. Explanation: $f''(x) = -\sin x \leq 0$ on $[0, \pi]$, so f is concave down and every chord lies below the graph. Correct statement: T_n overestimates when $f'' > 0$ (concave up) and underestimates when $f'' < 0$ (concave down).

Step 5. Contrast: $f(x) = x^2$ on $[0, 2]$ (Example 1) has $f'' = 2 > 0$ (concave up), which is why $T_4 = 2.75$ overestimated there.

Answer. False: $T_4 \approx 1.8961 < 2$ for $\sin x$ on $[0, \pi]$; the sign of the error is determined by the concavity (f'').